Hamilton decompositions of the directed 3-torus $D(m)$: explicit constructions for all $m \geq 3$

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Abstract

We study Hamilton decompositions of the directed three-dimensional torus

$$D(m) = \mathrm{Cay}((\mathbb{Z}_m)^3, \{e_1, e_2, e_3\}) \cong \vec{C}_m \square \vec{C}_m \square \vec{C}_m,$$

where $\vec{C}_m$ is the directed m-cycle. We prove that $D(m)$ admits a decomposition into three arc-disjoint directed Hamilton cycles for every $m \geq 3$.

For odd m, five Kempe swaps of the canonical arc-coloring produce the decomposition; explicit return-map formulas on a fixed plane show that each color class is Hamiltonian. For even m, a sign-product invariant yields a parity obstruction ruling out all Kempe-from-canonical approaches. We therefore introduce an explicit alternative direction assignment, Route E, for all even $m \geq 6$, whose return maps are derived in closed form. The boundary case $m = 4$ is handled separately.

Keywords
Cayley digraphs, directed Hamilton cycles, graph products, Hamilton decompositions, Kempe swaps

1 Introduction

Fix an integer $m \geq 3$ and write $\mathbb{Z}_m = \mathbb{Z}/m\mathbb{Z}$. Let $V = (\mathbb{Z}_m)^3$ and consider the directed graph $D(m)$ with arc set

$$(i, j, k) \to (i + 1, j, k), \qquad (i, j, k) \to (i, j + 1, k), \qquad (i, j, k) \to (i, j, k + 1),$$

with all arithmetic modulo m. Equivalently,

$$D(m) = \mathrm{Cay}((\mathbb{Z}_m)^3, \{e_1, e_2, e_3\}) \cong \vec{C}_m \square \vec{C}_m \square \vec{C}_m.$$

A *directed Hamilton decomposition* of $D(m)$ is a partition of its $3m^3$ arcs into three arc-disjoint directed Hamilton cycles.

Hamilton decompositions of Cartesian products of cycles have a long history. For undirected products, Kotzig proved that $C_m \square C_n$ decomposes into Hamilton cycles [9], and Foregger extended this to products of several cycles [8]. Broad surveys include Bermond [2] and Alspach, Bermond, and Sotteau [1]. For further results on Hamilton decompositions of graph products, see Stong [12].

On the directed side, Trotter and Erdős characterized when the product of two directed cycles is Hamiltonian [13], and Curran and Witte extended this by showing that the Cartesian product of any three or more nontrivial (length ≥ 2) directed cycles has a directed Hamilton cycle [6]. These results concern a single Hamilton cycle; the decomposition problem—partitioning all arcs into Hamilton

notation	meaning
$V = (\mathbb{Z}_m)^3$	vertex set of the directed torus $D(m)$
$S(i,j,k) = i + j + k \pmod m$	layer function; every legal arc increases S by 1
$P_t = \{S = t\}$	layer plane / return section

Table 1: Core geometric notation for $D(m)$.

cycles—was subsequently treated by Bogdanowicz [3, 4] and Darijani, Miraftab, and Morris [7]. As $D(m)$ is a Cayley digraph on an abelian group, our work also falls within the Cayley-graph Hamiltonicity tradition surveyed by Witte–Gallian [14] and Curran–Gallian [5]. Two closely related results are especially relevant. Bogdanowicz's decomposition results show that Cartesian products of directed cycles can be partitioned into cycles of a prescribed common length [3]; our problem is the extremal case where each cycle must be Hamiltonian. On the other hand, Darijani, Miraftab, and Morris show that for arc-disjoint Hamilton *paths*, the three-factor product remains an open case in the current general theory [7]. The symmetric three-dimensional torus $D(m)$ therefore sits at a natural boundary between existing Hamiltonicity results and full arc-decomposition. The present paper resolves that decomposition problem. The Kempe-swap recoloring operations central to our constructions are standard tools in edge-coloring and reconfiguration; see Mohar [10] and Stiebitz–Scheide–Toft–Favrholdt [11].

Our approach rests on three key ideas.

(i) A *sign-product invariant* for Kempe swaps, a general combinatorial identity holding for any coloring in which each color class is a permutation. From this invariant we derive a parity obstruction showing that the even case cannot be reached from the canonical arc-coloring by Kempe swaps alone.

(ii) A *return-map reduction*: because every arc of $D(m)$ increases the layer function $S = i + j + k$ by 1, Hamiltonicity on the full vertex set V of size m^3 reduces to the cycle structure of a return map on the plane $P_0 = \{S = 0\}$ of size m^2.

(iii) A necessary *case split*: the sign-product invariant forces the odd and even cases to use fundamentally different coloring strategies.

For odd m, the return maps arising from the Kempe-swap construction are affine skew-products whose single-cycle property follows from elementary number theory. For even m, the parity obstruction rules out all Kempe-from-canonical approaches, and the proof relies on an explicit direction assignment that breaks the canonical pattern on the layers $S \in \{0, 1, 2\}$, followed by a return-map analysis using three different first-return reductions.

Our main result is the following.

Theorem 1. *For every integer $m \geq 3$, the arc set of $D(m)$ decomposes into three directed Hamilton cycles.*

The proof splits into three cases: the odd case via an affine five-swap construction (Section 3), the even case $m \geq 6$ via Route E (Section 4; full return-map derivations in Appendices A–C), and the boundary case $m = 4$ via a finite witness (Appendix D).

2 Colorings, Kempe swaps, and return maps

2.1 Arc-colorings and direction assignments

We define a *coloring* of $D(m)$ to be a triple of permutations (f_0, f_1, f_2) on V such that

$$\{f_0(v), f_1(v), f_2(v)\} = \{v + e_1,\ v + e_2,\ v + e_3\} \qquad \text{for every } v \in V.$$

Equivalently, at each vertex this assigns a bijection from the color set $\{0, 1, 2\}$ to the three out-arcs $\{v \to v + e_1,\ v \to v + e_2,\ v \to v + e_3\}$. The color-$c$ subgraph is the functional digraph of f_c (i.e., the digraph with arc set $\{v \to f_c(v) : v \in V\}$) and is therefore a disjoint union of directed cycles. We say that color c is *Hamilton* if f_c is a single m^3-cycle.

Example 1 (Canonical coloring)**.** *The* canonical coloring *assigns the identity direction triple* $(d_0, d_1, d_2) = (0, 1, 2)$ *at every vertex, so that*

$$f_0(i, j, k) = (i + 1, j, k), \qquad f_1(i, j, k) = (i, j + 1, k), \qquad f_2(i, j, k) = (i, j, k + 1).$$

Each color is then a disjoint union of m^2 directed m-cycles. This coloring serves as the starting point for the odd-case construction (Section 3) and as the baseline whose sign product determines the parity barrier (Corollary 1).

We will also need a relaxed notion that drops the permutation requirement.

Definition 1 (Direction assignment)**.** *A* direction assignment *on $D(m)$ is a triple $\delta = (d_0, d_1, d_2)$ with $d_c : V \to \{0, 1, 2\}$ such that for every vertex v the triple $(d_0(v), d_1(v), d_2(v))$ is a permutation of $(0, 1, 2)$. It induces maps*

$$g_c(v) := v + e_{d_c(v)} \qquad (c \in \{0, 1, 2\}).$$

We call the direction assignment a coloring *precisely when each induced map g_c is a permutation of V.*

As we will see in Section 4, Route E will first be specified only as a direction assignment; its verification as a coloring is proved later from the bijectivity of its return maps via Lemma 4.

2.2 Kempe maps and Kempe swaps

Fix two colors $r \neq s$. Given a coloring (f_0, f_1, f_2), define the *Kempe map*

$$\tau_{r,s} := f_s^{-1} \circ f_r.$$

Its cycles are exactly the alternating (r, s)-colored cycles in the 2-colored subdigraph on colors r and s.

Definition 2 (Kempe swap)**.** *Let $X \subseteq V$ be a union of cycles of $\tau_{r,s}$. The* Kempe swap of colors (r, s) on X *is the operation producing a new triple (f'_0, f'_1, f'_2) by exchanging the color-r and color-s arcs at every vertex in X:*

$$f'_r(v) = \begin{cases} f_s(v), & v \in X, \\ f_r(v), & v \notin X, \end{cases} \qquad f'_s(v) = \begin{cases} f_r(v), & v \in X, \\ f_s(v), & v \notin X, \end{cases} \qquad f'_t = f_t \ (t \notin \{r, s\}).$$

Lemma 1 (Kempe swaps preserve validity)**.** *If (f_0, f_1, f_2) is a coloring and X is a union of cycles of $\tau_{r,s}$, then the Kempe swap on X produces another valid coloring.*

Proof. On each $\tau_{r,s}$-cycle $C \subseteq X$, the color-r and color-s arcs restrict to two perfect matchings from the source set C onto the common image set $f_r(C) = f_s(C)$. Indeed, if $C = (v_0, v_1, \ldots, v_{\ell-1})$, then $f_r(v_t) = f_s(v_{t+1})$ along the cycle. Swapping the two colors on C simply exchanges these matchings, so every vertex incident with C still has indegree 1 and outdegree 1 in each of the two colors. Outside X nothing changes. □

2.3 A parity barrier

For a permutation π on a finite set, let $\mathrm{sgn}(\pi) \in \{\pm 1\}$ denote its sign.

Theorem 2 (Sign-product invariant)**.** *Let (f_0, f_1, f_2) be a coloring and let (f'_0, f'_1, f'_2) be obtained by a Kempe swap between two colors. Then*

$$\mathrm{sgn}(f'_0)\,\mathrm{sgn}(f'_1)\,\mathrm{sgn}(f'_2) = \mathrm{sgn}(f_0)\,\mathrm{sgn}(f_1)\,\mathrm{sgn}(f_2).$$

Proof. It suffices to consider a swap between colors r and s, since the third color is unchanged. Because the swap support X is a disjoint union of $\tau_{r,s}$-cycles and the sign is multiplicative, it suffices to treat a single cycle. Let $C = (v_0, v_1, \ldots, v_{\ell-1})$ be one cycle of $\tau_{r,s} = f_s^{-1} \circ f_r$ contained in X. Define $\sigma \in S(V)$ to be the ℓ-cycle $(v_0\, v_1\, \cdots\, v_{\ell-1})$ on C and to fix $V \setminus C$ pointwise. A direct check using $f_r(v_t) = f_s(v_{t+1})$ shows

$$f'_r = f_r \circ \sigma^{-1}, \qquad f'_s = f_s \circ \sigma.$$

Therefore

$$\mathrm{sgn}(f'_r)\mathrm{sgn}(f'_s) = \mathrm{sgn}(f_r)\mathrm{sgn}(\sigma^{-1})\,\mathrm{sgn}(f_s)\mathrm{sgn}(\sigma) = \mathrm{sgn}(f_r)\mathrm{sgn}(f_s).$$

When X is a union of several cycles, applying the argument cycle by cycle shows that the swap on each cycle preserves the sign product, so the global sign product is invariant. □

Corollary 1 (Parity barrier for Kempe-from-canonical when m is even)**.** *Let m be even. Then no sequence of Kempe swaps starting from the canonical coloring can reach a Hamilton decomposition of $D(m)$.*

Proof. A directed Hamilton cycle on $N = m^3$ vertices is an N-cycle permutation and therefore has sign $(-1)^{N-1} = -1$ when m is even. Hence any decomposition into three Hamilton cycles must have sign product $(-1)^3 = -1$. In the canonical coloring, each color is a disjoint union of m^2 directed m-cycles. Each m-cycle has sign $(-1)^{m-1} = -1$ since m is even, and the product of m^2 such signs is $(-1)^{m^2} = +1$ since m^2 is even. Hence the total sign product is $(+1)^3 = +1$. The result now follows from Theorem 2. □

2.4 The layer function and return maps

Define the layer function

$$S(i, j, k) := i + j + k \pmod{m}.$$

Every allowed arc increases S by 1, so for every coloring and every color c one has $S(f_c(v)) = S(v) + 1$.

Lemma 2 (Plane invariance of Kempe maps)**.** *For any coloring and any two colors $r \neq s$, the Kempe map $\tau_{r,s} = f_s^{-1} \circ f_r$ preserves S. Consequently, every $\tau_{r,s}$-cycle is contained in a single plane $P_t := \{S = t\}$.*

Proof. Let $w = f_r(v)$. Then $S(w) = S(v) + 1$. If $u = \tau_{r,s}(v) = f_s^{-1}(w)$, then $f_s(u) = w$, so $S(w) = S(u) + 1$ and therefore $S(u) = S(v)$. □

Corollary 2 (Plane swaps are always Kempe-valid)**.** *For any $t \in \mathbb{Z}_m$, the plane P_t is a union of cycles of $\tau_{r,s}$ for every pair $r \neq s$. Hence swapping two colors on an entire plane is always a valid Kempe swap.*

Lemma 3 (Return maps control cycle structure)**.** *Fix a coloring and a color c. Let $P_0 = \{S = 0\}$ and let*

$$F_c := f_c^m|_{P_0} : P_0 \to P_0$$

be the return map. Then the directed cycles of f_c are in bijection with the directed cycles of F_c, and the length of a cycle of f_c is m times the length of the corresponding cycle of F_c. In particular, f_c is Hamilton on V if and only if F_c is a single m^2-cycle on P_0.

Proof. Along an f_c-orbit, the value of S increases by 1 at every step, so the orbit meets P_0 exactly once every m steps. The induced motion on P_0 is therefore exactly the return map $F_c = f_c^m|_{P_0}$. □

Lemma 4 (Validity via return maps)**.** *Let $f : V \to V$ be a directed color map such that every step increases the layer S by 1. Write $P_t = \{S = t\}$ and let $g_t : P_t \to P_{t+1}$ be the induced layer maps. If the return map*

$$R = g_{m-1} \circ \cdots \circ g_0 : P_0 \to P_0$$

is bijective, then each g_t is bijective. Consequently f is a permutation of V.

Proof. All layers have the same finite size m^2; this equal-size property is essential for the argument. Since R is bijective, it is injective. If g_0 were not injective, then R would not be injective. Hence g_0 is injective and therefore bijective.

Now write

$$g_{m-1} \circ \cdots \circ g_1 = R \circ g_0^{-1},$$

so the shorter composition is again bijective. Repeating the same argument inductively shows that every g_t is bijective. Therefore every vertex has indegree 1 and outdegree 1 in this color, so f is a permutation of V. □

3 An affine construction for odd m

3.1 The affine supports

Recall the canonical coloring from Example 1.

Let

$$P_0 := \{S = 0\}, \qquad P_1 := \{S = 1\},$$

and define the affine lines

$$L_0 := \{S = 0,\ k = 0\}, \qquad L_1 := \{S = 1,\ k = 0\}.$$

Theorem 3 (Affine five-swap construction)**.** *Start from the canonical coloring and perform the following five Kempe swaps:*

1. *swap colors $(0, 1)$ on the line L_0;*
2. *swap colors $(0, 2)$ on the plane P_0;*
3. *swap colors $(0, 1)$ on the plane P_0;*
4. *swap colors $(0, 1)$ on the line L_1;*

5. swap colors $(0,2)$ *on the plane* P_1.

This produces a valid coloring for every $m \geq 3$.

Proof of Theorem 3. Steps (2), (3), and (5) are Kempe-valid by Corollary 2 because they are supported on entire planes. For step (1), the coloring is canonical, and

$$\tau_{0,1} = f_1^{-1} \circ f_0 : (i,j,k) \mapsto (i+1, j-1, k)$$

preserves both S and k, acting on $L_0 = \{S = 0, k = 0\}$ as a single m-cycle. Hence (1) is Kempe-valid.

For step (4), note that steps (1)–(3) modify only tails in P_0. If $v \in P_1$, then $f_0(v)$ lies in P_2. Any color-1 arc landing in P_2 must originate in P_1, because every legal arc increases S by 1. Hence f_1^{-1} on the image of P_1 in P_2 depends only on the outgoing color-1 arcs from P_1, which remain canonical after steps (1)–(3). Therefore $\tau_{0,1}$ acts on P_1 as $(i,j,k) \mapsto (i+1, j-1, k)$, so L_1 is a single $\tau_{0,1}$-cycle and step (4) is Kempe-valid. □

Definition 3 (Bump maps)**.** *For* $v = (i,j,k) \in V$ *define*

$$\mathrm{bump}_i(v) = (i+1, j, k), \qquad \mathrm{bump}_j(v) = (i, j+1, k), \qquad \mathrm{bump}_k(v) = (i, j, k+1),$$

with all arithmetic modulo m.

Proposition 1 (Closed-form rule after the five swaps)**.** *Let* $m \geq 3$ *and let* $v = (i,j,k) \in V$ *with* $S = i + j + k \pmod{m}$. *After the five swaps of Theorem 3, the resulting maps* (f_0, f_1, f_2) *are*

$$f_0(v) = \begin{cases} \mathrm{bump}_j(v) & \text{if } S = 0 \text{ and } k \neq 0, \\ \mathrm{bump}_k(v) & \text{if } S = 1, \\ \mathrm{bump}_i(v) & \text{otherwise,} \end{cases}$$

$$f_1(v) = \begin{cases} \mathrm{bump}_k(v) & \text{if } S = 0, \\ \mathrm{bump}_i(v) & \text{if } S = 1 \text{ and } k = 0, \\ \mathrm{bump}_j(v) & \text{otherwise,} \end{cases}$$

$$f_2(v) = \begin{cases} \mathrm{bump}_j(v) & \text{if } S \in \{0,1\} \text{ and } k = 0, \\ \mathrm{bump}_i(v) & \text{if } S \in \{0,1\} \text{ and } k \neq 0, \\ \mathrm{bump}_k(v) & \text{otherwise.} \end{cases}$$

Proof. Outside $P_0 \cup P_1$ none of the five swaps acts, so the coloring remains canonical there.

On P_0, step (1) swaps colors $(0,1)$ on L_0 and does nothing on $P_0 \setminus L_0$. Thus after step (1) one has

$$(f_0, f_1, f_2) = \begin{cases} (\mathrm{bump}_j, \mathrm{bump}_i, \mathrm{bump}_k), & \text{on } L_0, \\ (\mathrm{bump}_i, \mathrm{bump}_j, \mathrm{bump}_k), & \text{on } P_0 \setminus L_0. \end{cases}$$

Step (2) swaps colors $(0,2)$ on all of P_0, so this becomes

$$(f_0, f_1, f_2) = \begin{cases} (\mathrm{bump}_k, \mathrm{bump}_i, \mathrm{bump}_j), & \text{on } L_0, \\ (\mathrm{bump}_k, \mathrm{bump}_j, \mathrm{bump}_i), & \text{on } P_0 \setminus L_0. \end{cases}$$

Step (3) swaps colors $(0,1)$ on all of P_0, yielding

$$(f_0, f_1, f_2) = \begin{cases} (\mathrm{bump}_i, \mathrm{bump}_k, \mathrm{bump}_j), & \text{on } L_0, \\ (\mathrm{bump}_j, \mathrm{bump}_k, \mathrm{bump}_i), & \text{on } P_0 \setminus L_0. \end{cases}$$

Since $L_0 = \{S = 0, k = 0\}$, this is exactly the stated rule on $S = 0$.

On P_1, the first three steps do nothing. Step (4) swaps colors $(0, 1)$ on L_1, so after step (4)

$$(f_0, f_1, f_2) = \begin{cases} (\text{bump}_j, \text{bump}_i, \text{bump}_k), & \text{on } L_1, \\ (\text{bump}_i, \text{bump}_j, \text{bump}_k), & \text{on } P_1 \setminus L_1. \end{cases}$$

Step (5) swaps colors $(0, 2)$ on all of P_1, so finally

$$(f_0, f_1, f_2) = \begin{cases} (\text{bump}_k, \text{bump}_i, \text{bump}_j), & \text{on } L_1, \\ (\text{bump}_k, \text{bump}_j, \text{bump}_i), & \text{on } P_1 \setminus L_1. \end{cases}$$

Since $L_1 = \{S = 1, k = 0\}$, this is exactly the stated rule on $S = 1$. □

Theorem 4 (Odd case)**.** *If m is odd, the coloring of Proposition 1 decomposes the arc set of $D(m)$ into three directed Hamilton cycles.*

Proof. By Lemma 3 it suffices to show that each return map $F_c = f_c^m|_{P_0}$ is a single m^2-cycle on P_0. Parameterize P_0 by $(i, k) \in (\mathbb{Z}_m)^2$ with $j \equiv -i - k$. A direct step-count from Proposition 1 gives

$$F_0(i, k) = (i - 2 + \mathbf{1}_{k=0},\, k + 1), \qquad F_1(i, k) = (i + \mathbf{1}_{k=-1},\, k + 1),$$

$$F_2(i, k) = (i + 2 - 2\mathbf{1}_{k=0},\, k - 2),$$

with all arithmetic modulo m.

For F_0, the second coordinate cycles by $k \mapsto k + 1$ and the net i-increment over one full k-cycle is

$$(-1) + (m - 1)(-2) = -2m + 1 \equiv 1 \pmod{m}.$$

Hence F_0^m fixes k and increments i by 1, so F_0 is a single m^2-cycle.

For F_1, again $k \mapsto k + 1$ is an m-cycle, and exactly once per k-cycle one has $k = -1$, so F_1^m fixes k and increments i by 1. Hence F_1 is a single m^2-cycle.

For F_2, the second coordinate evolves by $k \mapsto k - 2$. Since m is odd, -2 generates $\mathbb{Z}_m$, so k cycles through all values and hits $k = 0$ exactly once per m steps. Over one full k-cycle the net i-increment is

$$2(m - 1) \equiv -2 \pmod{m},$$

so F_2^m fixes k and shifts i by -2, which has order m because $\gcd(2, m) = 1$. Hence F_2 is a single m^2-cycle. Applying Lemma 3 completes the proof. □

Proposition 2 (Failure of color 2 for even m)**.** *Let $m \geq 4$ be even and let (f_0, f_1, f_2) be the affine coloring of Proposition 1. Then f_0 and f_1 are Hamilton cycles, while f_2 decomposes into exactly $m + 2$ directed cycles.*

Remark 1. *The failure is confined to a single color and is caused by the fact that -2 is not a unit modulo an even m. This diagnosis motivates the even-case strategy of modifying only the low-layer directions that feed into the return map for color 2.*

Proof of Proposition 2. The formulas for f_0 and f_1 are unchanged from the odd case, so those two colors are still Hamilton. For f_2, the return map is

$$F_2(i, k) = (i + 2 - 2\mathbf{1}_{k=0},\, k - 2).$$

Because m is even, the update $k \mapsto k-2$ preserves the parity of k, so P_0 splits into the odd-k half and the even-k half.

On the odd-k half, the second coordinate runs through all odd residues with period $m/2$ and never hits 0. Hence over one full k-cycle the first coordinate gains $2 \cdot (m/2) = m \equiv 0$, so every orbit there has length $m/2$. Since the odd-k half has size $m^2/2$, it contributes

$$\frac{m^2/2}{m/2} = m$$

cycles.

On the even-k half, the second coordinate again has period $m/2$, but now it hits 0 exactly once per such period. Therefore over one full k-cycle the net change in i is

$$2\left(\frac{m}{2} - 1\right) = m - 2 \equiv -2 \pmod{m}.$$

So the induced return map after $m/2$ steps is translation by -2 on $\mathbb{Z}_m$, which has exactly $\gcd(m, 2) = 2$ cycles. Hence the even-k half contributes 2 cycles.

Altogether f_2 decomposes into $m+2$ directed cycles. □

4 A return-map design principle and Route E

We now turn to the even case. The parity barrier of Corollary 1 rules out all Kempe-from-canonical approaches, so we adopt a direct construction. The odd proof shows that Hamiltonicity can often be reduced to a skew-product return map on $P_0 \cong (\mathbb{Z}_m)^2$; we begin by stating the underlying criterion.

Lemma 5 (Skew-product return maps)**.** *Fix $m \geq 1$, a step $d \in \mathbb{Z}_m$ with $\gcd(d, m) = 1$, and a function $\alpha : \mathbb{Z}_m \to \mathbb{Z}_m$. Define*

$$F(i, k) := (i + \alpha(k),\ k + d).$$

Then F is a permutation and

$$F^m(i, k) = (i + \Delta,\ k), \qquad \Delta := \sum_{t \in \mathbb{Z}_m} \alpha(t).$$

Consequently, every orbit of F has length $m^2/\gcd(\Delta, m)$, so F is a single m^2-cycle iff $\gcd(\Delta, m) = 1$.

Proof. Since d is a unit, $k \mapsto k + d$ is an m-cycle and F is bijective with inverse $(i, k) \mapsto (i - \alpha(k - d), k - d)$. Iterating gives

$$F^t(i, k) = \Big(i + \sum_{r=0}^{t-1} \alpha(k + rd),\ k + td\Big).$$

At $t = m$ the second coordinate returns to k and the sum ranges over all residues of $\mathbb{Z}_m$, giving Δ. The orbit length follows from the order of translation by Δ on the first coordinate. □

For the affine odd construction, Lemma 5 explains exactly why color 2 breaks when m is even: the step in the second coordinate is -2, which is not a unit modulo an even m. This isolates the even design problem: one must alter only the low layers in order to replace the nonunit drift in color 2 by a primitive two-dimensional bulk displacement, while keeping the set of exceptional vertices finite and geometrically explicit.

The sign-product barrier shows that, for even m, no Kempe-from-canonical derivation can possibly work. Accordingly, we introduce the Route E *direction assignment* on the low layers $S \in \{0, 1, 2\}$. All higher layers remain canonical.

For $v = (i, j, k) \in V$, write $S(v) = i + j + k \pmod m$. A direction triple $(d_0, d_1, d_2) \in \{0, 1, 2\}^3$ means that color c uses the arc $v \mapsto v + e_{d_c}$. At this stage we specify only a direction assignment in the sense of Definition 1; the fact that the associated maps are permutations, and hence form a coloring, is proved later in Theorem 5.

Definition 4 (Explicit Route E direction assignment)**.** *Let $m \geq 6$ be even. Define a direction triple $\delta(v) = (d_0(v), d_1(v), d_2(v))$ as follows. The layer-by-layer structure is summarized below; the formal rules appear in the subsequent paragraphs.*

layer	*rule*	*complexity*
$S \geq 3$	*canonical* $(0, 1, 2)$	*none*
$S = 2$	*two branches, split on* $j = 0$ *vs.* $j \neq 0$	2 *triples*
$S = 1$	*two branches, split on* $i = 0$ *vs.* $i \neq 0$	2 *triples*
$S = 0$	*six branches, depending on* $m \bmod 6$	≤ 6 *triples*

The design principle is as follows. Layers $S \geq 3$ keep the canonical triple, so the return map on P_0 sees their contribution as a uniform translation. Layers 1 and 2 introduce one branch each: they test whether a single coordinate vanishes after the preceding step, and swap two directions if so. Layer 0 carries the main splicing burden: its casework, which depends on $m \bmod 6$, arranges the finitely many defect points so that the resulting lane families are joined into a single cyclic order.

Layers $S \notin \{0, 1, 2\}$. *Use the canonical triple $(d_0, d_1, d_2) = (0, 1, 2)$.*

Layer $S = 1$. *Use*

$$(d_0, d_1, d_2) = \begin{cases} (1, 0, 2), & i = 0, \\ (2, 0, 1), & i \neq 0. \end{cases}$$

Layer $S = 2$. *Use*

$$(d_0, d_1, d_2) = \begin{cases} (2, 1, 0), & j = 0, \\ (0, 1, 2), & j \neq 0. \end{cases}$$

Layer $S = 0$, case $m \equiv 0, 2 \pmod 6$. *Use*

$$(d_0, d_1, d_2) = \begin{cases} (1, 0, 2), & v \in X_{102}, \\ (0, 2, 1), & v \in X_{021}, \\ (2, 1, 0), & v \in X_{210}, \\ (0, 1, 2), & v = (m-2, 1, 1), \\ (2, 0, 1), & v = (m-2, 2, 0), \\ (1, 2, 0), & \textit{otherwise}. \end{cases}$$

where the subscript on each set records the assigned direction triple:

$$\begin{aligned} X_{102} &:= \{(0, 0, 0)\} \cup \{(i, 1, m-1-i) : 1 \leq i \leq m-3\} \cup \{(m-1, 2, m-1)\}, \\ X_{021} &:= \{(0, 1, m-1)\} \cup \{(i, m-i, 0) : 1 \leq i \leq m-3\} \cup \{(m-1, 0, 1)\}, \\ X_{210} &:= \{(0, j, m-j) : 2 \leq j \leq m-1\} \cup \{(1, 0, m-1)\}. \end{aligned}$$

Layer $S = 0$, case $m \equiv 4 \pmod 6$. *Use*

$$(d_0, d_1, d_2) = \begin{cases} (1,0,2), & v \in Y_{102}, \\ (0,2,1), & v \in Y_{021}, \\ (2,1,0), & v \in Y_{210}, \\ (0,1,2), & v \in \{(1,1,m-2),(m-2,1,1)\}, \\ (2,0,1), & v \in \{(1,m-1,0),(m-2,2,0)\}, \\ (1,2,0), & \textit{otherwise.} \end{cases}$$

where

$$\begin{aligned} Y_{102} &:= \{(0,0,0)\} \cup \{(i,1,m-1-i) : 2 \le i \le m-3\} \cup \{(m-1,2,m-1)\}, \\ Y_{021} &:= \{(0,1,m-1)\} \cup \{(i,m-i,0) : 2 \le i \le m-3\} \cup \{(m-1,0,1)\}, \\ Y_{210} &:= \{(0,j,m-j) : 2 \le j \le m-1\} \cup \{(1,0,m-1)\} \\ &\qquad \cup \{(1,j,m-1-j) : 2 \le j \le m-2\} \cup \{(2,0,m-2),(2,m-1,m-1)\}. \end{aligned}$$

This completes the definition: set $f_c(v) = v + e_{d_c(v)}$; these are the associated color maps whose m-step behaviour on P_0 is analyzed below.

Once stated, the construction is verified purely combinatorially. We call a step at which the return map departs from the uniform translation a *stall*; the number of stalls per orbit is bounded independently of m.

Below we develop the return-map framework (§4.1), prove color 2 in full (§4.2), and assemble the main even-case theorem. Example 2 illustrates the full machinery on the smallest admissible case $m = 6$.

4.1 Framework for the Route E return maps

The appendices contain the detailed tables and the longer case-by-case bookkeeping, but the logical mechanism of the even proof can be stated in the main text. The first input is a three-step transducer extracted from Definition 4.

Proposition 3 (Three-step transducer for the Route E return maps)**.** *Parameterize*

$$P_0 = \{(i,j,k) \in \mathbb{Z}_m^3 : i + j + k \equiv 0\}$$

by

$$v(i,k) := (i, -i-k, k), \qquad (i,k) \in \mathbb{Z}_m^2.$$

For a color $c \in \{0,1,2\}$ and a start point $v(i,k) \in P_0$, write the first three color-c directions on the layers $S = 0,1,2$ as

$$\omega_c(i,k) = (a_c, b_c, c_c) \in \{0,1,2\}^3.$$

Then the return maps satisfy

$$R_0(i,k) = \big(i + N_0(\omega_0) - 3,\ k + N_2(\omega_0)\big),$$

$$R_1(i,k) = \big(i + N_0(\omega_1),\ k + N_2(\omega_1)\big),$$

$$R_2(i,k) = \big(i + N_0(\omega_2),\ k + N_2(\omega_2) - 3\big),$$

where $N_r(\omega)$ counts the occurrences of direction r in the word ω. Moreover:

(i) *for color* 1*, one always has* $b_1 = 0$ *and* $c_1 = 1$*;*

(ii) *for colors* 0 *and* 2*, the layer-*1 *direction depends only on whether the post-step-*1 i*-coordinate equals* 0*, and the layer-*2 *direction depends only on whether the post-step-*2 j*-coordinate equals* 0*.*

Appendix B therefore reduces the derivation of the piecewise formulas for R_0, R_1, R_2 *to a finite table evaluation on the layer-*0 *partition.*

Proof. After the first three steps from P_0, the orbit lies on layer $S = 3$. All subsequent layers are canonical, so until the next return to P_0 the map follows direction c for exactly $m - 3$ further steps. This gives the displayed displacement formulas.

Now write $j = -i - k$ on P_0. After step 1, the i-coordinate is

$$i_1 = i + \mathbf{1}_{a_c=0}.$$

By Definition 4, on layer $S = 1$ the color-1 direction is always 0, the color-0 direction is 1 if $i_1 = 0$ and 2 otherwise, and the color-2 direction is 2 if $i_1 = 0$ and 1 otherwise. So part (i) is immediate, and for colors 0 and 2 the second letter b_c is determined by the single test $i_1 = 0$.

After step 2, the j-coordinate is

$$j_2 = j + \mathbf{1}_{a_c=1} + \mathbf{1}_{b_c=1}.$$

Again by Definition 4, on layer $S = 2$ the color-0 direction is 2 if $j_2 = 0$ and 0 otherwise, while the color-2 direction is 0 if $j_2 = 0$ and 2 otherwise. Thus for colors 0 and 2 the third letter c_c is determined by the single test $j_2 = 0$. The appendix tables simply evaluate these tests on the finite partition of P_0 induced by the layer-0 rule. □

Lemma 6 (Generic three-step words and bulk vectors)**.** *Consider the Route E direction assignment. Suppose a start point* $(i, k) \in P_0 \cong \mathbb{Z}_m^2$ *has the base layer-*0 *triple* $(1, 2, 0)$ *and that, along the first two steps of the orbit, neither test* $i_1 = 0$ *on layer* 1 *nor* $j_2 = 0$ *on layer* 2 *is triggered. Then the three low-layer words are constant:*

$$\omega_0(i, k) = 120, \qquad \omega_1(i, k) = 201, \qquad \omega_2(i, k) = 012.$$

Consequently the return maps follow the bulk branches

$$R_0(i, k) = (i - 2, k + 1), \qquad R_1(i, k) = (i + 1, k + 1), \qquad R_2(i, k) = (i + 1, k - 2).$$

Proof. The base layer-0 triple $(1, 2, 0)$ means that on layer 0 the colors $0, 1, 2$ use directions $1, 2, 0$, respectively. If the post-step-1 test $i_1 = 0$ is not triggered, then the layer-1 triple is the generic one $(2, 0, 1)$. If the post-step-2 test $j_2 = 0$ is not triggered, then the layer-2 triple is the generic one $(0, 1, 2)$. Reading off the color-c letters from these three generic triples gives the displayed words $120, 201, 012$. The displacement formulas now follow immediately from Proposition 3. □

Remark 2 (structure of the defect sets)**.** *Lemma 6 explains the bulk of the construction: almost everywhere the low-layer word is constant, so the return maps are translations by the three displayed bulk vectors. Non-generic behaviour can occur only for three reasons:*

(a) *the start point already lies on one of the special layer-*0 *defect lines or endpoint corrections;*

(b) *after the first step, the test* $i_1 = 0$ *is triggered on layer* 1*;*

(c) *after the second step, the test* $j_2 = 0$ *is triggered on layer* 2.

In the (i, k) *coordinates on* P_0*, each of these conditions is affine, which is why the defect support in Lemma 9 consists of affine lines plus finitely many isolated boundary points. The boundary-point exceptions are then tuned only to splice the long arithmetic* lane families*—maximal runs of consecutive lanes that share the same bulk itinerary—produced by the bulk vectors into a single cyclic order. The residue-class split* $m \equiv 0, 2 \pmod 6$ *versus* $m \equiv 4 \pmod 6$ *appears only at this final splicing stage: for* $m \equiv 0, 2 \pmod 6$ *the primary defect lines already connect the relevant lane families, whereas for* $m \equiv 4 \pmod 6$ *one additional affine splice family is needed for colors* 1 *and* 0.

The next lemma is the counting lemma used in all three color proofs.

Lemma 7 (First-return counting)**.** *Let* $F : X \to X$ *be a map on a finite set* $|X| = N$*, and let* $L \subseteq X$*. Assume that for every* $x \in L$ *the first return time*

$$\rho(x) = \min\{t \geq 1 : F^t(x) \in L\}$$

exists, and write

$$T(x) = F^{\rho(x)}(x) \in L$$

for the induced first-return map. If

(i) T *is a single* $|L|$*-cycle on* L*, and*

(ii) $\sum_{x \in L} \rho(x) = N$*,*

then F *is a single* N*-cycle on* X*.*

Remark 3. *Lemma 7 is a discrete analogue of the Poincaré first-return map in dynamical systems: it reduces the global cycle structure of* F *to the dynamics of the induced map on a cross-section. We use it as the common final step in all three Route E color proofs.*

Proof. Pick $x_0 \in L$. Because T is a cycle on all of L, the return points along the F-orbit of x_0 are

$$x_0,\ T(x_0),\ T^2(x_0),\ \ldots,\ T^{|L|-1}(x_0),\ x_0,$$

and every point of L appears exactly once before the sequence closes.

Set

$$N_0 := \sum_{r=0}^{|L|-1} \rho(T^r(x_0)).$$

By construction, after N_0 applications of F one returns to x_0. So the F-orbit of x_0 is a cycle of length N_0. The hypothesis gives

$$N_0 = \sum_{x \in L} \rho(x) = N.$$

Hence one F-cycle already has size $N = |X|$, so it contains all points of X. Thus F is a single N-cycle. □

Across the three colors we choose coordinates adapted to the *bulk branch* of the return map—the branch that applies at all but boundedly many points of P_0. In each coordinate system the generic step advances one coordinate (the *clock*) uniformly while preserving the other (the *lane*). The finitely many non-generic branches are the *stalls* introduced above: at a stall the lane shifts by a bounded amount, and the clock may gain or lose an extra unit. In the orbit displays below, $\xrightarrow{\text{bulk}}$ or

$\xrightarrow{\text{bulk}^t}$ denotes one or t consecutive generic steps; $\xrightarrow{+1}$ or $\xrightarrow{+2}$ denotes a stall that increases the lane by 1 or 2.

Since the number of stalls per orbit is bounded independently of m, the two-dimensional dynamics on P_0 reduce to one-dimensional first-return maps on explicit transversals. On those transversals, most lanes follow the same arithmetic rule; only $O(1)$ boundary lanes have different itineraries.

Lemma 8 (bulk coordinates for the three colors)**.** *Define linear maps $\Phi_c : P_0 \cong \mathbb{Z}_m^2 \to \mathbb{Z}_m^2$ by*

$$\Phi_1(i,k) = (i-k,\ k),$$

$$\Phi_2(i,k) = (2i+k,\ -i-k),$$

$$\Phi_0(i,k) = (i+2k,\ k).$$

Let

$$b_1 = (1,1), \qquad b_2 = (1,-2), \qquad b_0 = (-2,1).$$

Then each Φ_c is invertible over $\mathbb{Z}_m$, and in the corresponding (u,t)-coordinates the bulk move $z \mapsto z + b_c$ becomes

$$(u,t) \mapsto (u, t+1).$$

Proof. The matrices of Φ_1, Φ_2, Φ_0 are

$$M_1 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \qquad M_2 = \begin{pmatrix} 2 & 1 \\ -1 & -1 \end{pmatrix}, \qquad M_0 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

Their determinants are $1, -1, 1$, so each is a unit in $\mathbb{Z}_m$ and each map is invertible. A direct multiplication gives

$$M_1 b_1 = (0,1), \qquad M_2 b_2 = (0,1), \qquad M_0 b_0 = (0,1),$$

which is exactly the claim. □

Lemma 9 (finite-defect normal form for Route E)**.** *For each color $c \in \{0,1,2\}$, let*

$$\widehat{R}_c := \Phi_c \circ R_c \circ \Phi_c^{-1}.$$

Then there is a defect set $E_c \subseteq \mathbb{Z}_m^2$, contained in a finite union of affine lines together with finitely many isolated points, such that

$$\widehat{R}_c(u,t) = (u, t+1) \qquad \text{for } (u,t) \notin E_c.$$

More precisely:

(i) *For color 1, every defect step is one of*

$$(u,t) \mapsto (u+1, t), \qquad (u,t) \mapsto (u+2, t),$$

and the defect set is contained in

$$u + t = 0, \qquad u + 2t = m - 1,$$

and, when $m \equiv 4 \pmod 6$, also

$$u + t = 1,$$

together with finitely many isolated points.

(ii) *For color* 2*, every defect step is one of*

$$(u,t) \mapsto (u-1,t+2), \quad (u,t) \mapsto (u+1,t+1), \quad (u,t) \mapsto (u-1,t+1),$$

$$(u,t) \mapsto (u-2,t+2), \quad (u,t) \mapsto (u+1,t),$$

and the defect set is contained in

$$t=1, \qquad t=m-1, \qquad u+2t=0, \qquad u+t=m-1,$$

together with finitely many isolated points.

(iii) *For color* 0*, every defect step is one of*

$$(u,t) \mapsto (u-2,t), \quad (u,t) \mapsto (u+3,t+3), \quad (u,t) \mapsto (u+2,t+2),$$

$$(u,t) \mapsto (u-1,t), \quad (u,t) \mapsto (u+1,t+2), \quad (u,t) \mapsto (u+1,t+1),$$

and the defect set is contained in

$$t=0, \qquad u=t+1,$$

and, when $m \equiv 4 \pmod 6$*, also*

$$u = 1+2t,$$

together with finitely many isolated points.

Proof. We compute directly from the piecewise formulas for R_0, R_1, R_2 recorded in Appendix A.

For color 1, the generic branch is $(i,k) \mapsto (i+1,k+1)$, while the two exceptional branches are $(i,k) \mapsto (i+1,k)$ and $(i,k) \mapsto (i+2,k)$. Under $\Phi_1(i,k) = (u,t) = (i-k,k)$ these become

$$(i+1,k+1) \mapsto (u,t+1), \qquad (i+1,k) \mapsto (u+1,t), \qquad (i+2,k) \mapsto (u+2,t).$$

The supporting conditions in Appendix A are $i=0$, $i=1$, and $i+k \equiv m-1$, plus finitely many exceptional boundary points. These are exactly the affine conditions

$$i=0 \iff u+t=0, \qquad i=1 \iff u+t=1, \qquad i+k \equiv m-1 \iff u+2t=m-1.$$

This proves (i).

For color 2, the generic branch is $(i,k) \mapsto (i+1,k-2)$. The exceptional branches listed in Appendix A are

$$(i+1,k-3), \qquad (i+2,k-3), \qquad (i,k-1), \qquad (i,k-2), \qquad (i+1,k-1).$$

Under $\Phi_2(i,k) = (u,t) = (2i+k,-i-k)$ one obtains

$$(i+1,k-2) \mapsto (u,t+1),$$

$$(i+1,k-3) \mapsto (u-1,t+2), \qquad (i+2,k-3) \mapsto (u+1,t+1),$$

$$(i,k-1) \mapsto (u-1,t+1), \qquad (i,k-2) \mapsto (u-2,t+2), \qquad (i+1,k-1) \mapsto (u+1,t).$$

The supporting conditions in Appendix A are $i+k \equiv m-1$, $i+k \equiv 1$, $k=0$, and $i=m-1$, together with finitely many exceptional points. These become

$$i+k \equiv m-1 \iff t=1,$$

$$i+k\equiv 1 \iff t=m-1,$$
$$k=0 \iff u+2t=0,$$
$$i=m-1 \iff u+t=m-1.$$

This proves (ii).

For color 0, the generic branch is $(i,k)\mapsto(i-2,k+1)$. The exceptional branches listed in Appendix A are

$$(i-2,k),\qquad (i-3,k+3),\qquad (i-2,k+2),\qquad (i-1,k),\qquad (i-3,k+2),\qquad (i-1,k+1).$$

Under $\Phi_0(i,k)=(u,t)=(i+2k,k)$ these become

$$(i-2,k+1)\mapsto(u,t+1),$$
$$(i-2,k)\mapsto(u-2,t),\qquad (i-3,k+3)\mapsto(u+3,t+3),\qquad (i-2,k+2)\mapsto(u+2,t+2),$$
$$(i-1,k)\mapsto(u-1,t),\qquad (i-3,k+2)\mapsto(u+1,t+2),\qquad (i-1,k+1)\mapsto(u+1,t+1).$$

The supporting conditions in Appendix A are $k=0$, $i+k\equiv 1$, and, in Case II, $i=1$, together with finitely many exceptional points. These become

$$k=0 \iff t=0,$$
$$i+k\equiv 1 \iff u=t+1,$$
$$i=1 \iff u=1+2t.$$

This proves (iii). □

Remark 4 (origin of the Case I / II split)**.** *The distinction between Case I ($m\equiv 0,2 \pmod 6$) and Case II ($m\equiv 4 \pmod 6$) that recurs in the color-by-color proofs below originates here. When $m\equiv 0,2 \pmod 6$ certain defect lines are absent—for instance, the line $u+t=1$ in part* (i) *and the line $u=1+2t$ in part* (iii) *appear only in Case II—so the orbit encounters fewer defect components and the first-return formulas are simpler. When $m\equiv 4 \pmod 6$ these extra defect lines are present, which introduces additional stall patterns and accounts for the distinct case analysis.*

Lemma 10 (defect-itinerary reduction)**.** *Let $F:\mathbb{Z}_m^2\to\mathbb{Z}_m^2$ be a map of the form*

$$F(z)=\begin{cases} G(z):=(u,t+1), & z\notin E,\\ D_\alpha(z):=(u+\Delta_\alpha,\ t+\tau_\alpha), & z\in E_\alpha,\end{cases}$$

where $E=\bigsqcup_\alpha E_\alpha$ is a finite disjoint union of defect components and each D_α has fixed increment $(\Delta_\alpha,\tau_\alpha)$. Let

$$L:=\{(u,0):u\in\mathbb{Z}_m\}.$$

Assume every point of L has a first return to L. Then for every start $(u,0)\in L$, the orbit segment before first return has a unique factorization

$$G^{a_0}D_{\alpha_1}G^{a_1}D_{\alpha_2}\cdots D_{\alpha_r}G^{a_r},$$

where each $a_j\geq 0$ and every point of each generic block lies outside E. If

$$F^{\rho(u)}(u,0)=(T(u),0),$$

then

$$T(u)=u+\sum_{j=1}^{r}\Delta_{\alpha_j}\pmod m,\qquad \rho(u)=\sum_{j=0}^{r}a_j+r.$$

color	bulk coordinates (u,t)	bulk branch	defect support / transversal
1	$(i-k,\, k)$	$(u,t)\mapsto(u,t+1)$	defects on $u+t=0$, $u+2t=m-1$, and in Case II also $u+t=1$; transversal $t=0$
2	$(2i+k,\, -i-k)$	$(u,t)\mapsto(u,t+1)$	defects on $t=1$, $t=m-1$, $u+2t=0$, $u+t=m-1$; transversal $t=0$
0	$(i+2k,\, k)$	$(u,t)\mapsto(u,t+1)$	defects on $t=0$, $u=t+1$, and in Case II also $u=1+2t$; transversal $t=0$

Table 2: Route E at a glance: adapted bulk coordinates, generic branch, and affine defect support for the three colors. The main text works out color 2 explicitly, while the appendix computes the remaining mod-6-dependent first-return data on the listed transversals.

Proof. Because F is deterministic, once an orbit point lies outside E it is forced to follow the generic branch G until it either returns to L or first meets one of the defect components. Starting from $(u,0)$, let a_0 be the maximal integer such that

$$(u,0),\ G(u,0),\ \ldots,\ G^{a_0-1}(u,0)$$

all lie outside E. If $G^{a_0}(u,0)\in L$, then there are no defect steps and the claim is immediate. Otherwise $G^{a_0}(u,0)$ lies in a unique defect component E_{α_1}, so the next step is forced to be D_{α_1}. Repeating the same argument recursively gives the displayed factorization, and uniqueness follows because there is never any branch choice.

The generic branch G leaves the u-coordinate unchanged, while each defect step D_{α_j} changes u by exactly Δ_{α_j}. Therefore the net lane change from $(u,0)$ to $(T(u),0)$ is

$$\sum_{j=1}^{r}\Delta_{\alpha_j},$$

which proves the formula for $T(u)$. The total number of steps is the number of generic steps plus the number of defect steps, giving

$$\rho(u)=\sum_{j=0}^{r}a_j+r.$$

□

Combining Lemma 9 with Lemma 10 reduces each Route E cycle proof to a one-dimensional first-return problem on the transversal $L=\{(u,0):u\in\mathbb{Z}_m\}$: one lists the defect components encountered by the bulk ray from $(u,0)$ before first return, reads off the lane increment and return time, and checks that the induced map T is a single m-cycle with $\sum_u\rho(u)=m^2$.

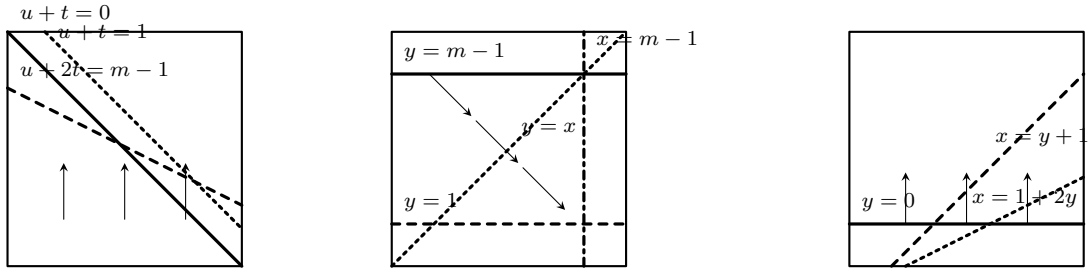
Remark 5 (R_2 is universal)**.** *The return-map formula for color* 2 *is the same for all even* $m\geq 6$*, independent of* $m \bmod 6$*; only colors* 0 *and* 1 *require a Case I/II split. For this reason, the main text works out color* 2 *completely before the appendix turns to the genuinely mod-*6*-dependent colors.*

Example 2 (Color 2 for $m=6$)**.** *We illustrate the return-map machinery on the smallest admissible even value* $m=6$*. In the working frame* $(x,y)=(i,\,i+k)$ *on* P_0*, the generic (bulk) step is* $G\colon(x,y)\mapsto(x+1,y-1)$*, and the transversal is* $L_2=\{(x,0):x\in\mathbb{Z}_6\}$*.*

Starting from lane $x=0$*:*

$$(0,0)\xrightarrow{E}(1,0).$$

This single defect step already returns to L_2*, giving* $T_2(0)=1$ *and* $\rho_2(0)=1$*.*



Color 1 in (u,t)-coordinates Color 2 in $(x,y)=(i,i+k)$ Color 0 in $(x,y)=(i+2k,k)$

Figure 1: Schematic defect geometry on P_0. The left and right panels use the bulk coordinates of Lemma 8; the center panel (color 2) uses the working frame $(x,y)=(i,\,i+k)$. In each panel the arrows indicate the generic bulk branch, while the solid/dashed/dotted lines represent the affine defect families from Lemma 9. The isolated boundary-point corrections sit on these lines and act as the splice points described in Remark 2.

Starting from lane $x=1$, the orbit crosses two defect lines before returning:

$$(1,0)\xrightarrow{G}(2,5)\xrightarrow{C}(2,4)\xrightarrow{G}(3,3)\xrightarrow{D}(3,1)\xrightarrow{B}(5,0).$$

The defect sequence is C (line $y=m-1$), then D (diagonal $y=x$), then B (line $y=1$). Return data: $T_2(1)=5$, $\rho_2(1)=5$.

For the remaining lanes, the same method gives $T_2(2)=0$ with $\rho_2(2)=12=2m$, and $T_2(x)=x-1$ with $\rho_2(x)=6=m$ for $x\in\{3,4,5\}$. The first-return map T_2 is therefore the 6-cycle

$$0\to1\to5\to4\to3\to2\to0,$$

and the return-time sum is $1+5+12+6+6+6=36=6^2=m^2$. Lemma 7 now implies that R_2 is a single 36-cycle on P_0.

4.2 The color-2 return map

We now carry out the color-2 computation in full. This is the cleanest representative case because its return map does not depend on $m \bmod 6$. For color 2 it is convenient to work directly in the coordinates $(x,y)=(i,\,i+k)$ rather than the bulk frame of Lemma 8; in this frame the generic branch becomes $G\colon (x,y)\mapsto(x+1,y-1)$ and the defect lines are those shown in the center panel of Figure 1.

Lemma 11. *In the coordinates $(x,y)=(i,i+k)$ on P_0, the associated Route E m-step map is*

$$R_2(x,y)=\begin{cases}(x+1,y-2), & (x,y)=(2,2),\\ (x+2,y-1), & y=1,\ (x,y)\notin\{(1,1),(m-1,1)\},\\ (x,y-1), & \big(y=m-1,\ x\neq m-1\big)\ \text{or}\ (x,y)=(m-1,m-2),\\ (x,y-2), & \big(y=x,\ x\notin\{0,2,m-1\}\big)\ \text{or}\ (x,y)=(m-1,0),\\ (x+1,y), & \big(x=m-1,\ y\notin\{0,m-2\}\big)\ \text{or}\ (x,y)=(0,0),\\ (x+1,y-1), & \text{otherwise.}\end{cases}$$

All arithmetic is modulo m.

Proof. Substitute $i = x$ and $k = y - x$ into the closed-form formula for R_2 from Appendix A and simplify each branch. The six displayed moves are exactly the transformed images of the six original branches. $\square$

For the orbit traces below write

$$G(x, y) = (x + 1, y - 1)$$

for the generic branch of Lemma 11. The other five displayed branches will be referred to as A, B, C, D, E in the order listed in the lemma: A applies at the isolated point $(2, 2)$; B is the defect on the line $y = 1$; C is the defect on $y = m - 1$; D is the defect on $y = x$; E is the defect on $x = m - 1$.

The transversal is

$$L_2 = \{(x, 0) : x \in \mathbb{Z}_m\}.$$

Proposition 4. *For every even $m \geq 6$, the first-return map on L_2 is*

$$T_2(x) = \begin{cases} 1, & x = 0, \\ m - 1, & x = 1, \\ 0, & x = 2, \\ x - 1, & 3 \leq x \leq m - 1, \end{cases}$$

with return times

$$\rho_2(x) = \begin{cases} 1, & x = 0, \\ m - 1, & x = 1, \\ 2m, & x = 2, \\ m, & 3 \leq x \leq m - 1. \end{cases}$$

Proof. We treat the boundary values $x \in \{0, 1, 2, 4, m - 2, m - 1\}$ individually and then give the generic odd-x and even-x formulas. Fix $x \in \mathbb{Z}_m$ and start from $(x, 0) \in L_2$.

Case $x = 0$. The point $(0, 0)$ is branch E, so

$$(0, 0) \xrightarrow{E} (1, 0).$$

Hence $T_2(0) = 1$ and $\rho_2(0) = 1$.

Case $x = 1$. The orbit is

$$(1, 0) \xrightarrow{G} (2, m-1) \xrightarrow{C} (2, m-2) \xrightarrow{G^{\frac{m}{2}-2}} \left(\frac{m}{2}, \frac{m}{2}\right) \xrightarrow{D} \left(\frac{m}{2}, \frac{m}{2} - 2\right) \xrightarrow{G^{\frac{m}{2}-3}} (m-3, 1) \xrightarrow{B} (m-1, 0).$$

Between the displayed defect points, no other special branch can occur: before reaching $\left(\frac{m}{2}, \frac{m}{2}\right)$ the orbit has $x < m - 1$, $y \notin \{0, 1, m - 1\}$, and meets the diagonal only at the displayed point; after the D-move the next special point is $(m - 3, 1)$. Therefore

$$T_2(1) = m - 1, \qquad \rho_2(1) = 1 + 1 + \left(\frac{m}{2} - 2\right) + 1 + \left(\frac{m}{2} - 3\right) + 1 = m - 1.$$

Case $x = 2$. The orbit is

$$(2, 0) \xrightarrow{G} (3, m-1) \xrightarrow{C} (3, m-2) \xrightarrow{G^{m-4}} (m-1, 2) \xrightarrow{E} (0, 2) \xrightarrow{G} (1, 1) \xrightarrow{D} (1, m-1) \xrightarrow{C} (1, m-2) \xrightarrow{G^{m-3}} (m-2, 1) \xrightarrow{B} ($$

Again, every point between the displayed defects is generic: before $(m-1,2)$ the orbit avoids the diagonal and the line $y=1$; after $(0,2)$ the only diagonal hit is $(1,1)$; after $(1,m-2)$ the next defect is $(m-2,1)$. Hence

$$T_2(2)=0, \qquad \rho_2(2)=1+1+(m-4)+1+1+1+1+(m-3)+1=2m.$$

Case x odd, $3 \le x \le m-3$. Set

$$d=\frac{x+m-1}{2}.$$

Then the orbit is

$$(x,0) \xrightarrow{G} (x{+}1,m{-}1) \xrightarrow{C} (x{+}1,m{-}2) \xrightarrow{G^{\frac{m-x-3}{2}}} (d,d) \xrightarrow{D} (d,d{-}2) \xrightarrow{G^{\frac{m-x-1}{2}}} (m{-}1,x{-}2) \xrightarrow{E} (0,x{-}2) \xrightarrow{G^{x-3}} (x{-}3,1$$

The first generic block runs on the line of ordinary sum $x+m-1$, which meets the diagonal exactly at (d,d). After the D-move the orbit runs on the line of ordinary sum $x+m-3$, and its next defect is $(m-1,x-2)$. After the E-move the orbit runs on the line of ordinary sum $x-2$, whose first defect is $(x-3,1)$. Thus

$$T_2(x)=x-1,$$

and

$$\rho_2(x)=1+1+\frac{m-x-3}{2}+1+\frac{m-x-1}{2}+1+(x-3)+1=m.$$

Case $x=4$. Here the orbit is

$$(4,0) \xrightarrow{G} (5,m-1) \xrightarrow{C} (5,m-2) \xrightarrow{G^{m-6}} (m-1,4) \xrightarrow{E} (0,4) \xrightarrow{G^2} (2,2) \xrightarrow{A} (3,0).$$

The displayed formula also covers the border case $m=6$, where the block G^{m-6} is empty. Hence

$$T_2(4)=3, \qquad \rho_2(4)=1+1+(m-6)+1+2+1=m.$$

Case x even, $6 \le x \le m-4$. Set

$$e=\frac{x}{2}.$$

Then the orbit is

$$(x,0) \xrightarrow{G} (x{+}1,m{-}1) \xrightarrow{C} (x{+}1,m{-}2) \xrightarrow{G^{m-x-2}} (m{-}1,x) \xrightarrow{E} (0,x) \xrightarrow{G^e} (e,e) \xrightarrow{D} (e,e{-}2) \xrightarrow{G^{e-3}} (x{-}3,1) \xrightarrow{B} (x{-}1,0$$

Before $(m-1,x)$ there is no diagonal hit because the first generic block lies on a line of odd ordinary sum. After the E-move the orbit lies on the line of ordinary sum x, whose first diagonal point is (e,e). After the D-move the orbit lies on the line of ordinary sum $x-2$, whose next defect is $(x-3,1)$. Therefore

$$T_2(x)=x-1,$$

and

$$\rho_2(x)=1+1+(m-x-2)+1+e+1+(e-3)+1=m.$$

Case $x=m-2$ (so necessarily $m \ge 8$). The orbit is

$$(m{-}2,0) \xrightarrow{G} (m{-}1,m{-}1) \xrightarrow{E} (0,m{-}1) \xrightarrow{C} (0,m{-}2) \xrightarrow{G^{\frac{m-2}{2}}} \left(\frac{m-2}{2},\frac{m-2}{2}\right) \xrightarrow{D} \left(\frac{m-2}{2},\frac{m-6}{2}\right) \xrightarrow{G^{\frac{m-8}{2}}} (m{-}5,1$$

Hence

$$T_2(m-2) = m-3, \qquad \rho_2(m-2) = 1+1+1+\frac{m-2}{2}+1+\frac{m-8}{2}+1 = m.$$

Case $x = m-1$**.** The orbit is

$$(m-1,0) \xrightarrow{D} (m-1,m-2) \xrightarrow{C} (m-1,m-3) \xrightarrow{E} (0,m-3) \xrightarrow{G^{m-4}} (m-4,1) \xrightarrow{B} (m-2,0).$$

So

$$T_2(m-1) = m-2, \qquad \rho_2(m-1) = 1+1+1+(m-4)+1 = m.$$

The seven cases exhaust $x \in \mathbb{Z}_m$, and the displayed formulas follow. □

Corollary 3 (color 2 is Hamilton for Route E)**.** *For every even $m \geq 6$, the associated map R_2 is a single m^2-cycle on P_0.*

Proof. By Proposition 4, the induced lane map follows the cyclic order

$$(0,1 \mid m-1, m-2, \ldots, 2),$$

namely $T_2(0) = 1$, $T_2(1) = m-1$, $T_2(2) = 0$, and $T_2(x) = x-1$ for $3 \leq x \leq m-1$. Hence T_2 is a single m-cycle. Also

$$\sum_{x \in \mathbb{Z}_m} \rho_2(x) = 1 + (m-1) + 2m + (m-3)m = m^2.$$

Lemma 7 now implies that R_2 is a single m^2-cycle. □

Theorem 5 (Route E for even $m \geq 6$)**.** *For every even integer $m \geq 6$, the Route E direction assignment (Definition 4) is a valid coloring of $D(m)$, and its three color classes are directed Hamilton cycles.*

Proof. Proposition 3 reduces the derivation of the return maps to a finite low-layer table, and Appendix B carries out that evaluation, yielding the closed-form formulas for R_0, R_1, R_2 recorded in Appendix A. Lemmas 8, 9, and 10 show that each return map is a finite-defect skew product whose cycle structure is governed by a one-dimensional first-return map on a transversal. Corollary 3 proves in the main text that R_2 is a single m^2-cycle on P_0. The color-2 argument serves as the template for the remaining two colors: in each case, one passes to adapted bulk coordinates (Lemma 8), identifies the affine defect lines (Lemma 9), and reads off the induced lane map from the defect itinerary (Lemma 10). Colors 0 and 1 differ from color 2 only in their bulk coordinate systems, their defect geometry, and the resulting mod-6 case split. Appendix C carries out this analysis: Corollary 6 concludes that R_1 is a single m^2-cycle, and Corollary 9 does the same for R_0. Lemma 7 therefore implies that each R_c is a single m^2-cycle on P_0. In particular, each R_c is bijective, so Lemma 4 implies that the Route E direction assignment is a valid coloring. Then Lemma 3 shows that each color map is a directed Hamilton cycle on V. Because every Route E direction triple is a permutation of $(0,1,2)$, the three Hamilton cycles are arc-disjoint and together cover all arcs of $D(m)$. □

Remark 6 (Finite-defect design principle)**.** *The Route E construction shares the same return-map framework as the odd affine construction—Hamiltonicity of the full color map on V is reduced to the cycle structure of a return map on P_0, which is further reduced to a one-dimensional first-return problem—but the execution is considerably more intricate. Where the odd case enjoys a single uniform return-map formula, the even return maps have piecewise definitions with an $m \bmod 6$ case split and boundary-point exceptions. Its low-layer modifications are supported on a bounded number of affine lines and exceptional points inside the planes $S = 0,1,2$. Away from this defect set, the return maps behave like simple bulk translations on P_0.*

5 The finite case $m = 4$

Route E is designed for $m \geq 6$; the case $m = 4$ is handled by a separate finite witness. Appendix D prints the complete direction table, from which the three Hamilton cycles can be read off by finite iteration; a machine-readable supplement records the orbits explicitly.

Proposition 5 (The finite case $m = 4$)**.** *The graph $D(4)$ admits a decomposition into three directed Hamilton cycles.*

Proof. Appendix D records a complete direction table for a coloring of $D(4)$. Starting from that table, each color orbit is obtained by finite iteration from any chosen initial vertex; in the accompanying machine-readable supplement `m4_witness.json` we record the three orbits starting from $(0, 0, 0)$, and the script `verify_m4_witness.py` checks that each one has length 64 and agrees with the displayed table. Thus the finite verification splits into two transparent steps: the table shows that every vertex sees a permutation of the three allowed directions, hence defines a valid direction assignment, and the extracted orbit data show that each color map is a single 64-cycle on V, hence a bijection; together these establish a valid coloring. Therefore the three color classes are Hamilton cycles and form a Hamilton decomposition of $D(4)$. □

6 Completion for all $m \geq 3$

Proof of Theorem 1. If m is odd, apply Theorem 4. If $m = 4$, apply Proposition 5. If m is even and $m \geq 6$, apply Theorem 5. These cases exhaust all integers $m \geq 3$. □

7 Discussion

The odd and even constructions are fundamentally different. For odd m, five Kempe swaps from the canonical coloring yield affine skew-product return maps whose single-cycle property follows from elementary modular arithmetic. For even m, the sign-product invariant rules out any Kempe-from-canonical approach, and the proof instead relies on the explicit Route E direction assignment, whose modifications are concentrated on the three lowest layers. The infinite-family proofs are purely combinatorial; the boundary case $m = 4$ is certified by an explicit finite witness (Appendix D).

Several natural questions remain. First, it would be desirable to unify the Route E case analysis—currently split by the residue of m modulo 6—into a single conceptual argument. At the level of the induced lane maps, the present proof shows that the low-layer defects have a single role: they splice a bounded number of bulk arithmetic lane families into one cyclic order. A shorter derivation of those splice rules, or a structural explanation of why the mod-6 split is unavoidable, would both be welcome. Second, the parity-obstruction viewpoint motivates the search for analogous invariants and obstruction-breaking constructions in higher-dimensional directed tori $\vec{C}_m^d$ and in broader families of Cayley digraphs. In particular, does Corollary 1 generalize to higher dimensions, and if so, does the obstruction admit a similar low-layer fix?

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Conflict of interest

The author declares no conflict of interest.

Data availability statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study. Supporting verification code and the machine-readable $m = 4$ witness referenced in Appendix E are supplied with the submission as supporting information.

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A Route E return maps

Definition 4 appears in Section 4. This appendix records the closed-form return maps on P_0 derived from that definition in Appendix B and analyzed in Appendix C.

Return maps on P_0. Parameterize $P_0 = \{S = 0\}$ by

$$v(i,k) := (i, -i-k, k), \qquad (i,k) \in \mathbb{Z}_m^2.$$

For the explicit Route E direction assignment, define $R_c = f_c^m|_{P_0}$. (We write R_c rather than F_c to distinguish the associated m-step maps of Route E from the return maps F_c of the Kempe-swap coloring analyzed in Section 3.) The following piecewise formulas are derived directly from Definition 4 in Appendix B.

Case I: $m \equiv 0, 2 \pmod 6$**.**

$$R_0(i,k) = \begin{cases} (i-2,\ k) & (i,k) = (0,0), \\ (i-3,\ k+3) & (i,k) = (1, m-1), \\ (i-2,\ k+2) & (i,k) = (m-2, 0), \\ (i-1,\ k) & (i,k) = (m-1, 1), \\ (i-3,\ k+2) & i+k \equiv 1 \text{ and } (i,k) \neq (1,0), \\ (i-1,\ k+1) & ((k=0 \text{ and } 1 \le i \le m-3) \text{ or } (i,k) \in \{(0,m-1),(m-2,1)\}), \\ (i-2,\ k+1) & \text{otherwise.} \end{cases}$$

$$R_1(i,k) = \begin{cases} (i+2,\ k) & \big((i+k \equiv m-1 \text{ and } 1 \le i \le m-3) \\ & \quad \text{or } (i,k) \in \{(0,0),(m-2,0),(m-1,m-1)\}\big), \\ (i+1,\ k) & \big((i=0 \text{ and } 1 \le k \le m-2) \\ & \quad \text{or } (i,k) \in \{(1,m-1),(m-2,1)\}\big), \\ (i+1,\ k+1) & \text{otherwise.} \end{cases}$$

Case II: $m \equiv 4 \pmod 6$**.**

$$R_0(i,k) = \begin{cases} (i-2,\ k) & (i,k) = (0,0), \\ (i-1,\ k) & (i,k) = (m-1, 1), \\ (i-3,\ k+3) & (i,k) \in \{(1,m-1),(2,m-2)\}, \\ (i-2,\ k+2) & \big((i=1 \text{ and } 0 \le k \le m-3) \text{ or } (i,k) \in \{(2,m-1),(m-2,0)\}\big), \\ (i-3,\ k+2) & i+k \equiv 1 \text{ and } (i,k) \notin \{(1,0),(2,m-1)\}, \\ (i-1,\ k+1) & \big((k=0 \text{ and } 2 \le i \le m-3) \\ & \quad \text{or } (i,k) \in \{(0,m-1),(1,m-2),(m-2,1)\}\big), \\ (i-2,\ k+1) & \text{otherwise.} \end{cases}$$

$$R_1(i,k) = \begin{cases} (i+2,\ k) & ((i+k \equiv m-1 \text{ and } 2 \le i \le m-3) \\ & \quad \text{or } (i,k) \in \{(0,0),(1,0),(m-2,0),(m-1,m-1)\}), \\ (i+1,\ k) & ((i=0 \text{ and } 1 \le k \le m-2) \text{ or } (i=1 \text{ and } 1 \le k \le m-1) \\ & \quad \text{or } (i,k) \in \{(2,m-2),(2,m-1),(m-2,1)\}), \\ (i+1,\ k+1) & \text{otherwise.} \end{cases}$$

Uniform for all even $m \ge 6$.

$$R_2(i,k) = \begin{cases} (i+1,\ k-3) & (i,k) = (2,0), \\ (i+2,\ k-3) & i+k \equiv 1 \text{ and } (i,k) \notin \{(1,0),(m-1,2)\}, \\ (i,\ k-1) & ((k \equiv m-1-i \text{ and } i \not\equiv m-1) \text{ or } (i,k) = (m-1,m-1)), \\ (i,\ k-2) & ((k=0 \text{ and } i \notin \{0,2,m-1\}) \text{ or } (i,k) = (m-1,1)), \\ (i+1,\ k-1) & ((i=m-1 \text{ and } k \notin \{1,m-1\}) \text{ or } (i,k) = (0,0)), \\ (i+1,\ k-2) & \text{otherwise.} \end{cases}$$

B Deriving the Route E return maps from Definition 4

In this appendix we derive the return maps recorded in Appendix A directly from the low-layer Route E rule. The argument does not appeal to the defect-cycle structure; it tracks only the first three layers $S = 0, 1, 2$.

Proposition 6. *For every even $m \ge 6$, the return maps R_0, R_1, R_2 induced by Definition 4 are exactly the piecewise formulas displayed in Appendix A.*

Proof. Parameterize

$$P_0 = \{(i,j,k) \in \mathbb{Z}_m^3 : i+j+k \equiv 0\}$$

by

$$v(i,k) := (i, -i-k, k), \qquad (i,k) \in \mathbb{Z}_m^2.$$

Fix a color $c \in \{0,1,2\}$ and start from $v(i,k) \in P_0$. Write the first three color-c directions encountered on the layers $S = 0, 1, 2$ as

$$\omega_c(i,k) = (a_c, b_c, c_c) \in \{0,1,2\}^3.$$

After those three steps the orbit lies on layer $S = 3$. From there until the next return to P_0 all layers are canonical, so the remaining $m-3$ steps are all in direction c. Hence the whole return map is determined by the low-layer word $\omega_c(i,k)$.

If $N_r(\omega)$ denotes the number of occurrences of direction r in a word ω, then

$$R_0(i,k) = \big(i + N_0(\omega_0) - 3,\ k + N_2(\omega_0)\big),$$

$$R_1(i,k) = \big(i + N_0(\omega_1),\ k + N_2(\omega_1)\big),$$

$$R_2(i,k) = \big(i + N_0(\omega_2),\ k + N_2(\omega_2) - 3\big).$$

So it remains only to determine the low-layer words.

From Definition 4:

- on layer $S = 1$, the triple is $(1,0,2)$ if $i = 0$, and $(2,0,1)$ if $i \ne 0$;

- on layer $S = 2$, the triple is $(2, 1, 0)$ if $j = 0$, and $(0, 1, 2)$ if $j \neq 0$.

Therefore:

- for color 0, the layer-1 direction depends on whether the post-step-1 i-coordinate equals 0, and the layer-2 direction depends on whether the post-step-2 j-coordinate equals 0;
- for color 1, the directions on layers 1 and 2 are always 0 and 1;
- for color 2, the layer-1 direction depends on whether the post-step-1 i-coordinate equals 0, and the layer-2 direction depends on whether the post-step-2 j-coordinate equals 0.

Case I: $m \equiv 0, 2 \pmod 6$**.** In the (i, k)-coordinates on P_0, the layer-0 direction triples become

$$\begin{aligned}
A &:= \{(0,0)\} \cup \{(i,k) : i + k \equiv m - 1,\ 1 \leq i \leq m - 3\} \cup \{(m-1, m-1)\}, \\
B &:= \{(0, m-1)\} \cup \{(i, 0) : 1 \leq i \leq m - 3\} \cup \{(m-1, 1)\}, \\
C &:= \{(0,k) : 1 \leq k \leq m - 2\} \cup \{(1, m-1)\}, \\
D &:= \{(m-2, 1)\}, \qquad E := \{(m-2, 0)\},
\end{aligned}$$

corresponding respectively to the triples $102, 021, 210, 012, 201$, while the remaining points

$$F := \mathbb{Z}_m^2 \setminus (A \cup B \cup C \cup D \cup E)$$

carry the triple 120.

Color 1**.** On layers 1 and 2, color 1 always uses directions 0 and 1. So the low-layer word is always of the form $(a, 0, 1)$, where a is the layer-0 color-1 direction. Thus:

- if $a = 0$, then $\omega_1 = 001$ and $R_1(i, k) = (i + 2, k)$;
- if $a = 1$, then $\omega_1 = 101$ and $R_1(i, k) = (i + 1, k)$;
- if $a = 2$, then $\omega_1 = 201$ and $R_1(i, k) = (i + 1, k + 1)$.

Reading off the sets on which the layer-0 color-1 direction equals 0, 1, or 2 gives exactly the Case I formula for R_1 in Appendix A.

Color 0**.** For color 0, we determine the word $\omega_0 = (a_0, b_0, c_0)$ algebraically. Step 1 goes in direction a_0, producing $i_1 = i + \mathbf{1}_{a_0=0}$. The layer-1 rule gives $b_0 = 1$ if $i_1 \equiv 0 \pmod m$, and 2 otherwise. Step 2 goes in direction b_0, producing

$$j_2 = j + \mathbf{1}_{a_0=1} + \mathbf{1}_{b_0=1}, \qquad j \equiv -i - k \pmod m.$$

The layer-2 rule gives $c_0 = 2$ if $j_2 \equiv 0 \pmod m$, and 0 otherwise. Table 3 executes this arithmetic trace on each region of P_0. Grouping the start points by the final displacement yields exactly the displayed Case I formula for R_0.

Table 3: Derivation of low-layer words ω_0 for color 0, Case I ($m \equiv 0, 2 \pmod 6$).

Set	Condition / point	a_0	$i_1 = i + \mathbf{1}_{a_0=0}$	b_0	$j_2 = j + \mathbf{1}_{a_0=1} + \mathbf{1}_{b_0=1}$	c_0	ω_0	Disp. $R_0(i,k)$
A	$(0,0)$	1	$0 \Rightarrow$	1	$0+1+1=2 \not\equiv 0 \Rightarrow$	0	110	$(i-2,k)$
	$A \setminus \{(0,0)\}$	1	$i \not\equiv 0 \Rightarrow$	2	$j+1+0 \not\equiv 0 \Rightarrow$	0	120	$(i-2,k+1)$
B	$(m-1,1)$	0	$0 \Rightarrow$	1	$0+0+1=1 \not\equiv 0 \Rightarrow$	0	010	$(i-1,k)$
	$B \setminus \{(m-1,1)\}$	0	$i+1 \not\equiv 0 \Rightarrow$	2	$j+0+0=j \not\equiv 0 \Rightarrow$	0	020	$(i-1,k+1)$
C	$(0,1)$	2	$0 \Rightarrow$	1	$-1+0+1=0 \Rightarrow$	2	212	$(i-3,k+2)$
	$(1,m-1)$	2	$1 \not\equiv 0 \Rightarrow$	2	$0+0+0=0 \Rightarrow$	2	222	$(i-3,k+3)$
	remaining C	2	$0 \Rightarrow$	1	$j+0+1 \not\equiv 0 \Rightarrow$	0	210	$(i-2,k+1)$
D	$(m-2,1)$	0	$m-1 \not\equiv 0 \Rightarrow$	2	$1+0+0=1 \not\equiv 0 \Rightarrow$	0	020	$(i-1,k+1)$
E	$(m-2,0)$	2	$m-2 \not\equiv 0 \Rightarrow$	2	$2+0+0=2 \not\equiv 0 \Rightarrow$	0	220	$(i-2,k+2)$
F	$i+k \equiv 1$ (equiv. $j \equiv -1$)	1	$i \not\equiv 0 \Rightarrow$	2	$-1+1+0=0 \Rightarrow$	2	122	$(i-3,k+2)$
	otherwise	1	$i \not\equiv 0 \Rightarrow$	2	$j+1+0 \not\equiv 0 \Rightarrow$	0	120	$(i-2,k+1)$

Color 2. Similarly, the layer-1 rule for color 2 gives $b_2 = 2$ if $i_1 \equiv 0 \pmod m$, and 1 otherwise. The layer-2 rule gives $c_2 = 0$ if $j_2 \equiv 0 \pmod m$, and 2 otherwise. Table 4 tracks this arithmetic. Grouping by displacement gives exactly the Case I formula for R_2.

Table 4: Derivation of low-layer words ω_2 for color 2, Case I ($m \equiv 0, 2 \pmod 6$).

Set	Condition / point	a_2	$i_1 = i + \mathbf{1}_{a_2=0}$	b_2	$j_2 = j + \mathbf{1}_{a_2=1} + \mathbf{1}_{b_2=1}$	c_2	ω_2	Disp. $R_2(i,k)$
A	$(0,0)$	2	$0 \Rightarrow$	2	$0+0+0=0 \Rightarrow$	0	220	$(i+1,k-1)$
	$A \setminus \{(0,0)\}$	2	$i \not\equiv 0 \Rightarrow$	1	$j+0+1 \not\equiv 0 \Rightarrow$	2	212	$(i,k-1)$
B	$(0,m-1)$	1	$0 \Rightarrow$	2	$1+1+0=2 \not\equiv 0 \Rightarrow$	2	122	$(i,k-1)$
	$(2,0)$	1	$2 \not\equiv 0 \Rightarrow$	1	$-2+1+1=0 \Rightarrow$	0	110	$(i+1,k-3)$
	remaining B	1	$i \not\equiv 0 \Rightarrow$	1	$j+1+1 \not\equiv 0 \Rightarrow$	2	112	$(i,k-2)$
C	$(0,1)$	0	$1 \not\equiv 0 \Rightarrow$	1	$-1+0+1=0 \Rightarrow$	0	010	$(i+2,k-3)$
	remaining C	0	$i+1 \not\equiv 0 \Rightarrow$	1	$j+0+1 \not\equiv 0 \Rightarrow$	2	012	$(i+1,k-2)$
D	$(m-2,1)$	2	$m-2 \not\equiv 0 \Rightarrow$	1	$1+0+1=2 \not\equiv 0 \Rightarrow$	2	212	$(i,k-1)$
E	$(m-2,0)$	1	$m-2 \not\equiv 0 \Rightarrow$	1	$2+1+1=4 \not\equiv 0 \Rightarrow$	2	112	$(i,k-2)$
F	$i=m-1$	0	$0 \Rightarrow$	2	$j+0+0=j \not\equiv 0 \Rightarrow$	2	022	$(i+1,k-1)$
	$i+k \equiv 1$ (equiv. $j \equiv -1$)	0	$i+1 \not\equiv 0 \Rightarrow$	1	$-1+0+1=0 \Rightarrow$	0	010	$(i+2,k-3)$
	otherwise	0	$i+1 \not\equiv 0 \Rightarrow$	1	$j+0+1 \not\equiv 0 \Rightarrow$	2	012	$(i+1,k-2)$

Case II: $m \equiv 4 \pmod 6$**.** In the (i,k)-coordinates on P_0, the layer-0 direction triples become

$$
\begin{aligned}
A &:= \{(0,0)\} \cup \{(i,k) : i+k \equiv m-1,\ 2 \le i \le m-3\} \cup \{(m-1,m-1)\},\\
B &:= \{(0,m-1)\} \cup \{(i,0) : 2 \le i \le m-3\} \cup \{(m-1,1)\},\\
C &:= \{(0,k) : 1 \le k \le m-2\} \cup \{(1,k) : 1 \le k \le m-3\} \cup \{(1,m-1)\} \cup \{(2,m-2),(2,m-1)\},\\
D &:= \{(1,m-2),(m-2,1)\}, \qquad E := \{(1,0),(m-2,0)\},
\end{aligned}
$$

corresponding again to the triples $102, 021, 210, 012, 201$, with 120 on the remaining points

$$F := \mathbb{Z}_m^2 \setminus (A \cup B \cup C \cup D \cup E).$$

Color 1. Exactly the same low-layer argument applies. In Case II the layer-0 color-1 direction equals 0 on

$$(i + k \equiv m - 1,\ 2 \le i \le m - 3) \text{ or } (i, k) \in \{(0, 0), (1, 0), (m - 2, 0), (m - 1, m - 1)\},$$

it equals 1 on

$$(i = 0,\ 1 \le k \le m - 2) \text{ or } (i = 1,\ 1 \le k \le m - 1) \text{ or } (i, k) \in \{(2, m - 2), (2, m - 1), (m - 2, 1)\},$$

and it equals 2 otherwise. This is exactly the Case II formula for R_1.

Color 0. Applying the same coordinate checks over the Case II partition generates Table 5. Using the identical word-to-displacement dictionary from Case I verifies the displayed Case II formula for R_0.

Table 5: Derivation of low-layer words ω_0 for color 0, Case II ($m \equiv 4 \pmod 6$).

Set	Condition / point	a_0	$i_1 = i + \mathbf{1}_{a_0=0}$	b_0	$j_2 = j + \mathbf{1}_{a_0=1} + \mathbf{1}_{b_0=1}$	c_0	ω_0	Disp. $R_0(i,k)$
A	$(0,0)$	1	$0 \Rightarrow$	1	$0+1+1=2 \not\equiv 0 \Rightarrow$	0	110	$(i-2,k)$
	$A \setminus \{(0,0)\}$	1	$i \not\equiv 0 \Rightarrow$	2	$j+1+0 \not\equiv 0 \Rightarrow$	0	120	$(i-2,k+1)$
B	$(m-1,1)$	0	$0 \Rightarrow$	1	$0+0+1=1 \not\equiv 0 \Rightarrow$	0	010	$(i-1,k)$
	$B \setminus \{(m-1,1)\}$	0	$i+1 \not\equiv 0 \Rightarrow$	2	$j+0+0=j \not\equiv 0 \Rightarrow$	0	020	$(i-1,k+1)$
C	$(0,1)$	2	$0 \Rightarrow$	1	$-1+0+1=0 \Rightarrow$	2	212	$(i-3,k+2)$
	$(1,m-1)$, $(2,m-2)$	2	$i \not\equiv 0 \Rightarrow$	2	$0+0+0=0 \Rightarrow$	2	222	$(i-3,k+3)$
	$(1,k)$ with $1 \le k \le m-3$, and $(2,m-1)$	2	$i \not\equiv 0 \Rightarrow$	2	$j+0+0=j \not\equiv 0 \Rightarrow$	0	220	$(i-2,k+2)$
	$(0,k)$ with $2 \le k \le m-2$	2	$0 \Rightarrow$	1	$j+0+1 \not\equiv 0 \Rightarrow$	0	210	$(i-2,k+1)$
D	$(1,m-2)$, $(m-2,1)$	0	$i+1 \not\equiv 0 \Rightarrow$	2	$1+0+0=1 \not\equiv 0 \Rightarrow$	0	020	$(i-1,k+1)$
E	$(1,0)$, $(m-2,0)$	2	$i \not\equiv 0 \Rightarrow$	2	$j+0+0=j \not\equiv 0 \Rightarrow$	0	220	$(i-2,k+2)$
F	$i+k \equiv 1$ (equiv. $j \equiv -1$)	1	$i \not\equiv 0 \Rightarrow$	2	$-1+1+0=0 \Rightarrow$	2	122	$(i-3,k+2)$
	otherwise	1	$i \not\equiv 0 \Rightarrow$	2	$j+1+0 \not\equiv 0 \Rightarrow$	0	120	$(i-2,k+1)$

Color 2. The analogous evaluation for color 2 is tracked in Table 6. Grouping the outputs matches the uniform formula for R_2 displayed in Appendix A.

Table 6: Derivation of low-layer words ω_2 for color 2, Case II ($m \equiv 4 \pmod 6$).

Set	Condition / point	a_2	$i_1 = i + \mathbf{1}_{a_2=0}$	b_2	$j_2 = j + \mathbf{1}_{a_2=1} + \mathbf{1}_{b_2=1}$	c_2	ω_2	Disp. $R_2(i,k)$
A	$(0,0)$	2	$0 \Rightarrow$	2	$0+0+0=0 \Rightarrow$	0	220	$(i+1,k-1)$
	$A \setminus \{(0,0)\}$	2	$i \not\equiv 0 \Rightarrow$	1	$j+0+1 \not\equiv 0 \Rightarrow$	2	212	$(i,k-1)$
B	$(0,m-1)$	1	$0 \Rightarrow$	2	$1+1+0=2 \not\equiv 0 \Rightarrow$	2	122	$(i,k-1)$
	$(2,0)$	1	$2 \not\equiv 0 \Rightarrow$	1	$-2+1+1=0 \Rightarrow$	0	110	$(i+1,k-3)$
	remaining B	1	$i \not\equiv 0 \Rightarrow$	1	$j+1+1 \not\equiv 0 \Rightarrow$	2	112	$(i,k-2)$
C	$(0,1)$, $(2,m-1)$	0	$i+1 \not\equiv 0 \Rightarrow$	1	$-1+0+1=0 \Rightarrow$	0	010	$(i+2,k-3)$
	remaining C	0	$i+1 \not\equiv 0 \Rightarrow$	1	$j+0+1 \not\equiv 0 \Rightarrow$	2	012	$(i+1,k-2)$
D	$(1,m-2)$, $(m-2,1)$	2	$i \not\equiv 0 \Rightarrow$	1	$1+0+1=2 \not\equiv 0 \Rightarrow$	2	212	$(i,k-1)$
E	$(1,0)$, $(m-2,0)$	1	$i \not\equiv 0 \Rightarrow$	1	$j+1+1 \not\equiv 0 \Rightarrow$	2	112	$(i,k-2)$
F	$i = m-1$	0	$0 \Rightarrow$	2	$j+0+0=j \not\equiv 0 \Rightarrow$	2	022	$(i+1,k-1)$
	$i+k \equiv 1$ (equiv. $j \equiv -1$)	0	$i+1 \not\equiv 0 \Rightarrow$	1	$-1+0+1=0 \Rightarrow$	0	010	$(i+2,k-3)$
	otherwise	0	$i+1 \not\equiv 0 \Rightarrow$	1	$j+0+1 \not\equiv 0 \Rightarrow$	2	012	$(i+1,k-2)$

This proves all return-map formulas. □

C Hamiltonicity of the Route E return maps

C.1 Setup

The first-return framework of Section 4 applies uniformly to all three colors. The main text proves color 2 in full; this appendix records the remaining itinerary computations for colors 1 and 0. For color 1, Propositions 7 and 9 derive the first-return data; Corollary 6 establishes Hamiltonicity. For color 0, Propositions 11 and 13 do the same, with Corollary 9 completing the proof. The working coordinates and defect geometries for each color are summarized in Table 2.

The next lemma packages the cycle step that remains after the closed-form formulas for the first-return maps have been derived.

Lemma 12 (successor on cyclic blocks)**.** *Let X be a finite set written cyclically as*

$$X = (B_1 \mid B_2 \mid \cdots \mid B_r),$$

where each block $B_j = (x_{j,0}, x_{j,1}, \ldots, x_{j,\ell_j})$ is linearly ordered. Assume $T : X \to X$ satisfies

$$T(x_{j,s}) = x_{j,s+1} \quad (0 \le s < \ell_j), \qquad T(x_{j,\ell_j}) = x_{j+1,0},$$

with the block index j taken cyclically modulo r. Then T is a single $|X|$-cycle.

Proof. The hypotheses say exactly that T moves forward by one step in the cyclic concatenation

$$(x_{1,0}, x_{1,1}, \ldots, x_{1,\ell_1}, x_{2,0}, \ldots, x_{r,\ell_r}).$$

Hence every point lies on the same orbit, and that orbit has length $|X|$. □

In the propositions below we use Lemma 12 to repackage the induced lane maps T_c as successor permutations on cyclic concatenations of a bounded number of long arithmetic runs. This does

not replace the derivation of the closed-form formulas for T_c and ρ_c or the return-time sums; it only explains why the cycle structure is so simple once those formulas are known. When one of the displayed arithmetic runs is empty for the smallest admissible values of m, it is omitted from the cyclic concatenation.

C.2 Color 2

The color-2 return map is proved in full in the main text (Section 4.2). Proposition 4 derives the first-return data, and Corollary 3 establishes that R_2 is a single m^2-cycle for all even $m \geq 6$. We do not repeat those orbit traces here.

C.3 Color 1

By Lemma 8, the bulk coordinates for color 1 are $\Phi_1(i,k) = (u,t) = (i-k,\, k)$, in which the generic branch is $(u,t) \mapsto (u, t+1)$. By Lemma 9(i), the defect set lies on the affine lines $u+t=0$ and $u+2t = m-1$ (and, in Case II, also $u+t=1$), together with finitely many isolated points. Following the same pattern, it suffices to list the defect components met before first return and sum the lane increments.

We use a first-return analysis on the transversal

$$L_1 = \{(x,0) : x \in \mathbb{Z}_m\} \subseteq P_0.$$

Let $\rho_1(x)$ be the first return time of R_1 from $(x,0)$ to L_1, and define

$$R_1^{\rho_1(x)}(x,0) = (T_1(x), 0).$$

In the original (i,k)-coordinates, $u = i-k$ is the lane coordinate. A bulk move $(i,k) \mapsto (i+1, k+1)$ preserves u. The two defect branches act as stalls: $(i,k) \mapsto (i+1,k)$ increases u by 1 (defect line $u+t=0$, i.e. $i=0$), and $(i,k) \mapsto (i+2,k)$ increases u by 2 (defect line $u+2t = m-1$, i.e. diagonal $i+k = m-1$). Starting from $(x,0)$, the net change in i at first return is exactly the total lane increment contributed by the stalls.

C.3.1 Case I: $m \equiv 0, 2 \pmod 6$

Proposition 7. *Assume $m \equiv 0, 2 \pmod 6$. Then the first-return map and return times on L_1 are*

$$T_1(x) = \begin{cases} 2, & x = 0, \\ x+3, & 1 \leq x \leq m-4, \\ 1, & x = m-3, \\ 0, & x = m-2, \\ 3, & x = m-1, \end{cases} \qquad \rho_1(x) = \begin{cases} 1, & x = 0 \text{ or } x = m-2, \\ m+2, & 1 \leq x \leq m-4, \\ m+3, & x = m-3 \text{ or } x = m-1. \end{cases}$$

Proof. The closed-form map R_1 has three types of moves:

- bulk $(i,k) \mapsto (i+1, k+1)$,
- $+1$ stalls on $i=0$ together with $(1, m-1)$ and $(m-2, 1)$,
- $+2$ stalls on the diagonal $i+k = m-1$ together with $(0,0)$, $(m-2,0)$ and $(m-1, m-1)$.

The immediate cases are read directly from the formula:

$$(0,0) \mapsto (2,0), \qquad (m-2,0) \mapsto (0,0).$$

Hence

$$T_1(0) = 2, \quad \rho_1(0) = 1, \qquad T_1(m-2) = 0, \quad \rho_1(m-2) = 1.$$

Now fix $1 \le x \le m-4$.

Odd x. On the lane $u = x$ one reaches the diagonal $i + k = m - 1$ once before k wraps, namely at

$$k_1 = \frac{m-1-x}{2}, \qquad (i,k) = \left(\frac{m-1+x}{2}, \frac{m-1-x}{2}\right).$$

That point is in the $+2$ branch, so the lane becomes $x+2$. On the new lane $u = x+2$, the next defect before wrap is the vertical point $(0, m-x-2)$, which is a $+1$ stall, so the lane then becomes $x+3$. On that lane there is no diagonal defect before wrap, because the system

$$i - k = x + 3, \qquad i + k = m - 1$$

would force

$$k = \frac{m-x-4}{2},$$

which is not an integer when x is odd. Its vertical defect is at height $m-x-3$, already behind the current value $k = m-x-2$. Thus there are exactly two stalls, with total lane increment $2+1=3$. Therefore

$$T_1(x) = x+3, \qquad \rho_1(x) = m+2.$$

Even x. The lane $u = x$ has no diagonal hit. Its first defect is the vertical point $(0, m-x)$, which contributes a $+1$ stall and moves the orbit to lane $x+1$. That new lane is odd, so before wrap it hits the diagonal once, at

$$k_2 = m - \frac{x}{2} - 1,$$

and that diagonal hit is a $+2$ stall. So the lane becomes $x+3$. On lane $x+3$ the vertical defect is at height $m-x-3$, while the diagonal equation

$$i - k = x + 3, \qquad i + k = m - 1$$

gives the candidate height

$$k = \frac{m-x-4}{2}.$$

Both heights are strictly below the current value $k_2 = m - \frac{x}{2} - 1$. Hence there are no more stalls before return. Again the total lane increment is $1+2=3$, and the number of stalls is 2, giving

$$T_1(x) = x+3, \qquad \rho_1(x) = m+2.$$

It remains to treat the two boundary values.

The point $x = m-3$. The orbit begins

$$(m-3,0) \to (m-2,1) \to (m-1,1) \to (0,2) \to (1,2).$$

Thus it first gets a $+1$ stall at $(m-2,1)$ and then another $+1$ stall at $(0,2)$. At that point the orbit lies on the lane $u = i - k \equiv m-1$. The unique diagonal defect on this lane before k wraps is

$$\left(\frac{m}{2} - 1, \frac{m}{2}\right),$$

so the orbit continues

$$(1,2) \xrightarrow{\text{bulk } \frac{m}{2}-2} \left(\frac{m}{2}-1, \frac{m}{2}\right) \xrightarrow{+2} \left(\frac{m}{2}+1, \frac{m}{2}\right) \xrightarrow{\text{bulk } \frac{m}{2}} (1,0).$$

Hence the third stall is this +2 diagonal stall, there are exactly three stalls in total, and therefore

$$T_1(m-3) = 1, \qquad \rho_1(m-3) = m+3.$$

The point $x = m-1$. The orbit begins

$$(m-1,0) \to (0,1) \to (1,1),$$

so it gets a +1 stall at $(0,1)$. Later it reaches $(m-1, m-1)$, giving a +2 stall, and then $(1, m-1)$, giving the final +1 stall. At that point the orbit is on lane 3 with current clock value $k = m-1$. The vertical defect on lane 3 is at height $m-3$, while a diagonal defect on lane 3 would require

$$k = \frac{m-4}{2},$$

so both possible defect heights lie strictly behind the current value $m-1$. Hence there are no more defects before return. Therefore the total lane increment is 4 and the number of stalls is 3, so

$$T_1(m-1) = 3, \qquad \rho_1(m-1) = m+3.$$

This proves the stated formulas. □

Proposition 8 (splicing of arithmetic runs for color 1, Case I)**.** *Assume $m \equiv 0, 2 \pmod 6$. Then T_1 is the successor permutation on a cyclic order obtained from three long arithmetic runs together with $O(1)$ bridge vertices. More precisely,*

$$(0, 2, 5, \ldots, m-3 \mid 1, 4, 7, \ldots, m-1 \mid 3, 6, 9, \ldots, m-2)$$

when $m \equiv 2 \pmod 6$, and

$$(0, 2, 5, \ldots, m-1 \mid 3, 6, 9, \ldots, m-3, 1 \mid 4, 7, 10, \ldots, m-2)$$

when $m \equiv 0 \pmod 6$. Consequently T_1 is a single m-cycle.

Proof. By Proposition 7,

$$T_1(0) = 2, \qquad T_1(x) = x+3 \quad (1 \le x \le m-4), \qquad T_1(m-3) = 1, \qquad T_1(m-2) = 0, \qquad T_1(m-1) = 3.$$

Hence the generic action is translation by +3, and only the endpoint values $m-3, m-2, m-1$ alter the itinerary. If $m \equiv 2 \pmod 6$, the three +3-runs are

$$2, 5, \ldots, m-3, \qquad 1, 4, 7, \ldots, m-1, \qquad 3, 6, 9, \ldots, m-2,$$

with the special values splicing them in the order

$$0 \to 2, \qquad m-3 \to 1, \qquad m-1 \to 3, \qquad m-2 \to 0.$$

If $m \equiv 0 \pmod 6$, the same generic rule produces the runs

$$2, 5, \ldots, m-1, \qquad 3, 6, 9, \ldots, m-3, \qquad 1, 4, 7, \ldots, m-2,$$

and the endpoint values splice them in the order

$$0 \to 2, \qquad m-1 \to 3, \qquad m-3 \to 1, \qquad m-2 \to 0.$$

In either congruence class this is exactly the displayed cyclic order, so Lemma 12 applies. □

Corollary 4. *If $m \equiv 0, 2 \pmod 6$, then R_1 is a single m^2-cycle.*

Proof. Proposition 8 shows that T_1 is a single m-cycle. Also,

$$\sum_{x \in \mathbb{Z}_m} \rho_1(x) = 2 + (m-4)(m+2) + 2(m+3) = m^2.$$

Now apply Lemma 7. □

C.3.2 Case II: $m \equiv 4 \pmod 6$

Proposition 9. *Assume $m \equiv 4 \pmod 6$. Then the first-return map and return times on L_1 are*

$$T_1(x) = \begin{cases} 2, & x = 0, \\ 3, & x = 1, \\ 5, & x = 2, \\ x+2, & x \text{ even}, \ x \notin \{0, 2, m-2\}, \\ 0, & x = m-2, \\ x+6, & x \text{ odd}, \ x \notin \{1, m-3, m-1\}, \\ 4, & x = m-3, \\ 7, & x = m-1, \end{cases}$$

and

$$\rho_1(x) = \begin{cases} 1, & x \in \{0, 1, m-2\}, \\ m+3, & x = 2, \\ m+2, & x \text{ even}, \ x \notin \{0, 2, m-2\}, \\ m+4, & x \text{ odd}, \ x \notin \{1, m-3, m-1\}, \\ m+6, & x \in \{m-3, m-1\}. \end{cases}$$

Proof. In Case II the closed-form map has

- bulk $(i, k) \mapsto (i+1, k+1)$,
- $+1$ stalls on the lines $i = 0$ and $i = 1$, together with the special points $(2, m-2)$, $(2, m-1)$, $(m-2, 1)$,
- $+2$ stalls on the diagonal $i + k = m - 1$ for $2 \le i \le m-3$, together with the special points $(0, 0)$, $(1, 0)$, $(m-2, 0)$, $(m-1, m-1)$.

The immediate cases are read directly from the formula:

$$(0,0) \mapsto (2,0), \qquad (1,0) \mapsto (3,0), \qquad (m-2,0) \mapsto (0,0),$$

so

$$T_1(0) = 2, \quad T_1(1) = 3, \quad T_1(m-2) = 0,$$

and all three return times equal 1.

For $x = 2$, the lane $u = 2$ has no diagonal defect. The orbit runs generically to $(0, m-2)$, then encounters three consecutive $+1$ stalls:

$$(2,0) \xrightarrow{\text{bulk}^{m-2}} (0, m-2) \xrightarrow{+1} (1, m-2) \xrightarrow{+1} (2, m-2) \xrightarrow{+1} (3, m-2).$$

From there two bulk moves return to L_1. Hence

$$T_1(2) = 5, \qquad \rho_1(2) = m + 3.$$

Now let x be even with $4 \le x \le m - 4$. The lane $u = x$ has no diagonal defect. Its first defect is $(0, m - x)$, followed immediately by the $i = 1$ defect at the same height:

$$(x, 0) \xrightarrow{\text{bulk}^{m-x}} (0, m - x) \xrightarrow{+1} (1, m - x) \xrightarrow{+1} (2, m - x).$$

Because $x \neq 2$, the point $(2, m - x)$ is not one of the special $+1$ points. The orbit is now on the even lane $x + 2$. On that lane there is no diagonal defect before wrap, and the remaining vertical defects occur at heights $m - x - 2$ and $m - x - 1$, both already behind the current value $k = m - x$. So no further defect occurs before the return to L_1, and therefore

$$T_1(x) = x + 2, \qquad \rho_1(x) = m + 2.$$

Next let x be odd with $3 \le x \le m - 7$. Define

$$d_x = \frac{m - 1 - x}{2}, \qquad e_x = m - x - 2, \qquad f_x = m - \frac{x + 5}{2}.$$

Because x is odd, the first diagonal defect on lane $u = x$ occurs at height d_x, namely at

$$D_1 = \left(\frac{m - 1 + x}{2}, \frac{m - 1 - x}{2} \right).$$

So the orbit first performs one $+2$ stall there. It then lies on lane $x + 2$, whose next defects are the vertical points

$$V_0 = (0, e_x), \qquad V_1 = (1, e_x),$$

which contribute two consecutive $+1$ stalls. The orbit then lies on the odd lane $x + 4$. On that lane the lower diagonal height is already behind the current clock value, while the upper diagonal height is exactly f_x. Thus the next defect is the diagonal point

$$D_2 = \left(\frac{x + 3}{2}, m - \frac{x + 5}{2} \right),$$

which contributes the second $+2$ stall. After this move the orbit lies on lane $x + 6$. On that lane the remaining vertical defect heights are $m - x - 6$ and $m - x - 5$, and the remaining diagonal height before wrap is $m - (x + 7)/2$; all of these lie behind the current value $k = f_x$. So no further defect occurs before the return to L_1. In total the orbit has two $+2$ stalls and two $+1$ stalls, hence total lane increment 6 and return time $m + 4$:

$$T_1(x) = x + 6, \qquad \rho_1(x) = m + 4.$$

For the endpoint $x = m - 5$, the same initial defect pattern occurs: one $+2$ stall on the first diagonal hit, then two consecutive $+1$ stalls on the lines $i = 0$ and $i = 1$, and then one more $+2$ stall on the second diagonal hit. After that second diagonal stall the orbit lies on lane 1 at height $m/2$. The lower diagonal point on lane 1 is already behind the current clock value, and the only remaining intersection with the diagonal before wrap is $(0, m - 1)$. This point is not a defect: it is excluded both from the $i = 0$ $+1$ branch and from the diagonal $+2$ branch. So the next step is the bulk move

$$(0, m - 1) \mapsto (1, 0),$$

which is the first return to L_1. Thus

$$T_1(m-5) = 1 = (m-5)+6, \qquad \rho_1(m-5) = m+4.$$

For $x = m-3$, the orbit is

$$\begin{aligned}(m-3,0) &\xrightarrow{\text{bulk}} (m-2,1) \xrightarrow{+1} (m-1,1) \xrightarrow{\text{bulk}} (0,2) \xrightarrow{+1} (1,2)\\ &\xrightarrow{+1} (2,2) \xrightarrow{\text{bulk}^{m-3}} (m-1,m-1) \xrightarrow{+2} (1,m-1)\\ &\xrightarrow{+1} (2,m-1) \xrightarrow{+1} (3,m-1) \xrightarrow{\text{bulk}} (4,0).\end{aligned}$$

So the lane increment is $1+1+1+2+1+1 = 7$ and the return time is $m+6$. Hence

$$T_1(m-3) = 4, \qquad \rho_1(m-3) = m+6.$$

For $x = m-1$, the orbit is

$$\begin{aligned}(m-1,0) &\xrightarrow{\text{bulk}} (0,1) \xrightarrow{+1} (1,1) \xrightarrow{+1} (2,1) \xrightarrow{\text{bulk}^{\frac{m}{2}-2}} \left(\frac{m}{2}, \frac{m}{2}-1\right)\\ &\xrightarrow{+2}\xrightarrow{\text{bulk}^{\frac{m}{2}-2}} (0,m-3) \xrightarrow{+1} (1,m-3) \xrightarrow{+1} (2,m-3)\\ &\xrightarrow{+2} (4,m-3) \xrightarrow{\text{bulk}^3} (7,0).\end{aligned}$$

Thus the orbit has four $+1$ stalls and two $+2$ stalls, for total lane increment 8 and return time $m+6$. Therefore

$$T_1(m-1) = 7, \qquad \rho_1(m-1) = m+6.$$

□

Proposition 10 (splicing of arithmetic runs for color 1, Case II)**.** *Assume $m \equiv 4 \pmod 6$. Then T_1 is the successor permutation on a cyclic order obtained from three long arithmetic runs together with $O(1)$ bridge vertices:*

$$(0,2,5,11,\ldots,m-5,1 \mid 3,9,15,\ldots,m-1,7,13,\ldots,m-3 \mid 4,6,8,\ldots,m-2).$$

Consequently T_1 is a single m-cycle.

Proof. Proposition 9 shows that, away from the boundary values

$$0,1,2,m-3,m-2,m-1,$$

the map acts by

$$x \mapsto x+2 \quad \text{on even residues}, \qquad x \mapsto x+6 \quad \text{on odd residues}.$$

The special values then splice these generic arithmetic runs as follows:

$$0 \to 2 \to 5,$$

after which the odd $+6$-run continues through

$$5,11,\ldots,m-5$$

and ends at

$$m-5 \to 1.$$

Next

$$1 \to 3,$$

after which the odd +6-rule produces the second odd run

$$3, 9, 15, \ldots, m-1, 7, 13, \ldots, m-3,$$

and the boundary value

$$m-3 \to 4$$

enters the even run

$$4, 6, 8, \ldots, m-2.$$

Finally

$$m-2 \to 0$$

closes the cycle. Thus T_1 is exactly the successor map on the displayed cyclic order, so Lemma 12 applies. □

Corollary 5. *If $m \equiv 4 \pmod 6$, then R_1 is a single m^2-cycle.*

Proof. Proposition 10 shows that T_1 is a single m-cycle. Also,

$$\sum_{x \in \mathbb{Z}_m} \rho_1(x) = 3 + (m+3) + \frac{m-6}{2}(m+2) + \frac{m-6}{2}(m+4) + 2(m+6) = m^2.$$

Therefore Lemma 7 applies. □

Corollary 6. *For every even $m \geq 6$, the return map R_1 is a single m^2-cycle on P_0.*

C.4 Color 0

By Lemma 8, the bulk coordinates for color 0 are $\Phi_0(i,k) = (u,t) = (i+2k,\, k)$, in which the generic branch is $(u,t) \mapsto (u, t+1)$. By Lemma 9(iii), the defect set lies on the lines $t = 0$ and $u = t+1$ (and, in Case II, also $u = 1+2t$), together with finitely many isolated points. Since the bulk frame already uses the convenient coordinates $(x,y) = (u,t) = (i+2k,\, k)$, we work directly in this frame.

The transversal is

$$L_0 = \{(x, 0) : x \in \mathbb{Z}_m\}.$$

The generic move freezes x and increments y, so y is literally a clock while the exceptional branches change only the lane coordinate x.

C.4.1 Case I: $m \equiv 0, 2 \pmod 6$

Lemma 13. *Assume $m \equiv 0, 2 \pmod 6$. In the coordinates $(x, y) = (i + 2k, k)$, the return map R_0 becomes*

$$R_0(x,y) = \begin{cases} (m-2, 0), & (x,y) = (0,0), \\ (2,2), & (x,y) = (m-1, m-1), \\ (0,2), & (x,y) = (m-2, 0), \\ (0,1), & (x,y) = (1,1), \\ (x+1, y+2), & x = y+1,\ (x,y) \neq (1,0), \\ (x+1, y+1), & (y = 0,\ 1 \leq x \leq m-3) \text{ or } (x,y) \in \{(0,1), (m-2, m-1)\}, \\ (x, y+1), & \text{otherwise.} \end{cases}$$

All arithmetic is modulo m.

Proof. Substitute $i = x - 2y$ and $k = y$ into the Case I formula for R_0 in Appendix A and simplify branch by branch. The generic move $(i, k) \mapsto (i - 2, k + 1)$ becomes $(x, y) \mapsto (x, y + 1)$, and the six exceptional branches transform exactly as displayed. □

Write

$$G(x, y) = (x, y + 1)$$

for the generic branch in Lemma 13. The six exceptional moves will be denoted P, Q, R, S, A, B in the order displayed in the lemma: P and Q are isolated-point defects at $(0, 0)$ and $(m - 1, m - 1)$; R maps $(m - 2, 0)$ to $(0, 2)$; S maps $(1, 1)$ to $(0, 1)$; A is the defect on the line $u = t + 1$ (i.e. $x = y + 1$); B is the defect on the line $t = 0$ (i.e. $y = 0$) together with the boundary points $(0, 1)$ and $(m - 2, m - 1)$.

Proposition 11. *Assume $m \equiv 0, 2 \pmod 6$. Then the first-return map on L_0 is*

$$T_0(x) = \begin{cases} m-2, & x = 0, \\ x+2, & 1 \leq x \leq m-5, \\ m-1, & x = m-4, \\ 2, & x = m-3, \\ 1, & x = m-2, \\ 0, & x = m-1, \end{cases}$$

and the return times are

$$\rho_0(x) = \begin{cases} 1, & x = 0, \\ m-1, & 1 \leq x \leq m-4 \text{ or } x = m-1, \\ 2m-3, & x = m-3, \\ 2m-1, & x = m-2. \end{cases}$$

Proof. Fix $x \in \mathbb{Z}_m$ and start from $(x, 0) \in L_0$.

Case $x = 0$. By branch P,

$$(0,0) \xrightarrow{P} (m-2, 0),$$

so $T_0(0) = m - 2$ and $\rho_0(0) = 1$.

Case $1 \le x \le m-5$**.** The orbit is

$$(x,0) \xrightarrow{B} (x+1,1) \xrightarrow{G^{x-1}} (x+1,x) \xrightarrow{A} (x+2,x+2) \xrightarrow{G^{m-x-2}} (x+2,0).$$

Indeed, on the lane $x+1$ the first defect above $y=1$ is the A-point $(x+1,x)$. After that jump the orbit lies on lane $x+2$ at height $y=x+2$. On this lane the only possible A-point is $(x+2,x+1)$, whose height $x+1$ is already behind the current clock value. The special points P,Q,R,S lie on different lanes, and the B-branch occurs only at $y=0$ or at the isolated points $(0,1)$ and $(m-2,m-1)$, none of which lies ahead on lane $x+2$. Thus every remaining step before $y=0$ is generic. Therefore

$$T_0(x) = x+2, \qquad \rho_0(x) = 1+(x-1)+1+(m-x-2) = m-1.$$

Case $x = m-4$**.** Here

$$(m-4,0) \xrightarrow{B} (m-3,1) \xrightarrow{G^{m-5}} (m-3,m-4) \xrightarrow{A} (m-2,m-2) \xrightarrow{G} (m-2,m-1) \xrightarrow{B} (m-1,0).$$

So

$$T_0(m-4) = m-1, \qquad \rho_0(m-4) = 1+(m-5)+1+1+1 = m-1.$$

Case $x = m-3$**.** The orbit is

$$(m-3,0) \xrightarrow{B} (m-2,1) \xrightarrow{G^{m-4}} (m-2,m-3) \xrightarrow{A} (m-1,m-1) \xrightarrow{Q} (2,2) \xrightarrow{G^{m-2}} (2,0).$$

Hence

$$T_0(m-3) = 2, \qquad \rho_0(m-3) = 1+(m-4)+1+1+(m-2) = 2m-3.$$

Case $x = m-2$**.** The orbit is

$$(m-2,0) \xrightarrow{R} (0,2) \xrightarrow{G^{m-3}} (0,m-1) \xrightarrow{A} (1,1) \xrightarrow{S} (0,1) \xrightarrow{B} (1,2) \xrightarrow{G^{m-2}} (1,0).$$

Therefore

$$T_0(m-2) = 1, \qquad \rho_0(m-2) = 1+(m-3)+1+1+1+(m-2) = 2m-1.$$

Case $x = m-1$**.** The orbit is

$$(m-1,0) \xrightarrow{G^{m-2}} (m-1,m-2) \xrightarrow{A} (0,0).$$

So

$$T_0(m-1) = 0, \qquad \rho_0(m-1) = m-2+1 = m-1.$$

These cases exhaust $x \in \mathbb{Z}_m$. □

Proposition 12 (splicing of arithmetic runs for color 0, Case I)**.** *Assume* $m \equiv 0,2 \pmod 6$*. Then* T_0 *is the successor permutation on a cyclic order obtained from two long arithmetic runs together with* $O(1)$ *bridge vertices:*

$$(0, m-2 \mid 1,3,5,\dots,m-3 \mid 2,4,6,\dots,m-4 \mid m-1).$$

Consequently T_0 *is a single* m*-cycle.*

Proof. Proposition 11 gives

$$T_0(0) = m-2, \qquad T_0(x) = x+2 \quad (1 \le x \le m-5), \qquad T_0(m-4) = m-1,$$

$$T_0(m-3) = 2, \qquad T_0(m-2) = 1, \qquad T_0(m-1) = 0.$$

So the generic action is translation by $+2$ on the long odd and even runs

$$1, 3, 5, \ldots, m-3, \qquad 2, 4, 6, \ldots, m-4,$$

while the exceptional values splice them through the two-point block

$$0 \to m-2$$

and the one-point tail

$$m-1 \to 0.$$

More explicitly,

$$m-2 \to 1, \qquad m-3 \to 2, \qquad m-4 \to m-1, \qquad m-1 \to 0,$$

so T_0 is the successor map on the displayed cyclic order. Lemma 12 now applies. □

Corollary 7. *If $m \equiv 0, 2 \pmod 6$, then R_0 is a single m^2-cycle.*

Proof. Proposition 12 shows that T_0 is a single m-cycle. Also

$$\sum_{x \in \mathbb{Z}_m} \rho_0(x) = 1 + (m-3)(m-1) + (2m-3) + (2m-1) = m^2.$$

Lemma 7 now applies. □

C.4.2 Case II: $m \equiv 4 \pmod 6$

Lemma 14. *Assume $m \equiv 4 \pmod 6$. In the coordinates $(x, y) = (i + 2k, k)$, the return map R_0 becomes*

$$R_0(x,y) = \begin{cases} (m-2, 0), & (x,y) = (0,0), \\ (x+3, y+3), & (x,y) \in \{(m-1, m-1), (m-2, m-2)\}, \\ (x+2, y+2), & (x,y) = (m-2, 0) \text{ or } (x,y) = (0, m-1) \text{ or } x \equiv 1 + 2y \pmod m,\ 0 \le y \le m-3, \\ (0,1), & (x,y) = (1,1), \\ (x+1, y+2), & x = y+1,\ (x,y) \notin \{(1,0), (0, m-1)\}, \\ (x+1, y+1), & (y = 0,\ 2 \le x \le m-3) \text{ or } (x,y) \in \{(0,1), (m-3, m-2), (m-2, m-1)\}, \\ (x, y+1), & \text{otherwise.} \end{cases}$$

All arithmetic is modulo m.

Proof. Again substitute $i = x - 2y$ and $k = y$ into the Case II formula from Appendix A. The generic branch becomes $(x, y) \mapsto (x, y+1)$, and the exceptional branches simplify exactly as shown. □

As before write

$$G(x, y) = (x, y + 1)$$

for the generic move. For an odd lane $u \in \mathbb{Z}_m$ define the two R-heights

$$r_-(u) = \frac{u-1}{2}, \qquad r_+(u) = \frac{u-1}{2} + \frac{m}{2}.$$

These are exactly the two solutions $y \in \{0, \ldots, m-1\}$ of

$$u \equiv 1 + 2y \pmod{m}.$$

Proposition 13. *Assume $m \equiv 4 \pmod 6$. Then the first-return map on L_0 is given by*

$$T_0(0) = m-2, \qquad T_0(1) = 4, \qquad T_0(2) = 6, \qquad T_0(x) = x + 4 \quad (3 \le x \le m-7),$$

together with

$$T_0(m-6) = 3, \quad T_0(m-5) = m-1, \quad T_0(m-4) = 0, \quad T_0(m-3) = 2, \quad T_0(m-2) = 5, \quad T_0(m-1) = 1,$$

and the return times are

$$\rho_0(0) = 1, \qquad \rho_0(x) = m-2 \quad (\textit{for } 1 \le x \le m-7 \textit{ or } x = m-4),$$

$$\rho_0(m-6) = 2m-4, \quad \rho_0(m-2) = 2m-4, \quad \rho_0(m-5) = m-1, \quad \rho_0(m-1) = m-1, \quad \rho_0(m-3) = 2m-3.$$

Proof. Fix $x \in \mathbb{Z}_m$ and start from $(x, 0) \in L_0$.

Case $x = 0$. By the special point branch,

$$(0, 0) \xrightarrow{P} (m-2, 0),$$

so $T_0(0) = m-2$ and $\rho_0(0) = 1$.

Case $x = 1$. The orbit is

$$(1, 0) \xrightarrow{R} (3, 2) \xrightarrow{A} (4, 4) \xrightarrow{G^{m-4}} (4, 0).$$

So

$$T_0(1) = 4, \qquad \rho_0(1) = 1 + 1 + (m-4) = m-2.$$

Case $x = 2$. The orbit is

$$(2, 0) \xrightarrow{B} (3, 1) \xrightarrow{R} (5, 3) \xrightarrow{G} (5, 4) \xrightarrow{A} (6, 6) \xrightarrow{G^{m-6}} (6, 0),$$

hence

$$T_0(2) = 6, \qquad \rho_0(2) = 1 + 1 + 1 + 1 + (m-6) = m-2.$$

Case x odd, $3 \le x \le m-7$. The first move is

$$(x, 0) \xrightarrow{B} (x+1, 1),$$

and the lane $x + 1$ is even, so there is no R-branch on that lane. The next defect is the A-point $(x + 1, x)$. Thus

$$(x, 0) \xrightarrow{B} (x+1, 1) \xrightarrow{G^{x-1}} (x+1, x) \xrightarrow{A} (x+2, x+2).$$

Now the orbit lies on the odd lane $x+2$. Since the lower R-height

$$r_-(x+2) = \frac{x+1}{2}$$

is already behind the current value $y = x+2$, the next R-point is the upper one,

$$r_+(x+2) = \frac{x+1}{2} + \frac{m}{2}.$$

Because $x \le m-7$, this lies strictly between $x+2$ and m. So

$$(x+2, x+2) \xrightarrow{G^{\frac{m-x-3}{2}}} (x+2, r_+(x+2)) \xrightarrow{R} (x+4, r_+(x+2)+2)\,.$$

The new lane $x+4$ is even, so there is no further R-branch. Its only A-point is at height $x+3$, already behind the current height $r_+(x+2)+2$. The B-branch occurs only at $y=0$ or at the isolated points $(0,1)$, $(m-3, m-2)$, $(m-2, m-1)$, none of which lies ahead on lane $x+4$. Hence every remaining step before the return to L_0 is generic, and therefore

$$T_0(x) = x+4.$$

The time is

$$\rho_0(x) = 1 + (x-1) + 1 + \frac{m-x-3}{2} + 1 + \frac{m-x-5}{2} = m-2.$$

Case x even, $4 \le x \le m-8$. The orbit begins

$$(x, 0) \xrightarrow{B} (x+1, 1),$$

and now the lane $x+1$ is odd. Its next R-height is the lower one

$$r_-(x+1) = \frac{x}{2},$$

so

$$(x+1, 1) \xrightarrow{G^{\frac{x}{2}-1}} \left(x+1, \frac{x}{2}\right) \xrightarrow{R} \left(x+3, \frac{x}{2}+2\right).$$

The new lane $x+3$ is odd. Its lower R-height is $x/2+1$, already behind the current height, while the A-height is $x+2$ and, because $x \le m-8$, this comes before the upper R-height. Thus

$$\left(x+3, \frac{x}{2}+2\right) \xrightarrow{G^{\frac{x}{2}}} (x+3, x+2) \xrightarrow{A} (x+4, x+4).$$

The lane $x+4$ is even, so there is no R-branch on it. Its only A-point is at height $x+3$, already behind the current height $x+4$, and the B-branch occurs only at $y=0$ or at the isolated points $(0,1)$, $(m-3, m-2)$, $(m-2, m-1)$, none of which lies ahead on lane $x+4$. So

$$T_0(x) = x+4,$$

and

$$\rho_0(x) = 1 + \left(\frac{x}{2} - 1\right) + 1 + \frac{x}{2} + 1 + (m-x-4) = m-2.$$

Case $x = m-6$. Here the orbit is

$$(m-6, 0) \xrightarrow{B} (m-5, 1) \xrightarrow{G^{\frac{m-8}{2}}} \left(m-5, \frac{m-6}{2}\right) \xrightarrow{R} \left(m-3, \frac{m}{2}-1\right) \xrightarrow{G^{\frac{m-6}{2}}} (m-3, m-4) \xrightarrow{A} (m-2, m-2)$$

$$\xrightarrow{Q}(1,1)\xrightarrow{S}(0,1)\xrightarrow{B}(1,2)\xrightarrow{G^{\frac{m-4}{2}}}\left(1,\frac{m}{2}\right)\xrightarrow{R}\left(3,\frac{m}{2}+2\right)\xrightarrow{G^{\frac{m-4}{2}}}(3,0).$$

Thus

$$T_0(m-6)=3,$$

and

$$\rho_0(m-6)=1+\frac{m-8}{2}+1+\frac{m-6}{2}+1+1+1+1+\frac{m-4}{2}+1+\frac{m-4}{2}=2m-4.$$

Case $x=m-5$**.** The orbit is

$$(m-5,0)\xrightarrow{B}(m-4,1)\xrightarrow{G^{m-6}}(m-4,m-5)\xrightarrow{A}(m-3,m-3)\xrightarrow{G}(m-3,m-2)\xrightarrow{B}(m-2,m-1)\xrightarrow{B}(m-1,0).$$

Therefore

$$T_0(m-5)=m-1,\qquad \rho_0(m-5)=1+(m-6)+1+1+1+1=m-1.$$

Case $x=m-4$**.** The orbit is

$$(m-4,0)\xrightarrow{B}(m-3,1)\xrightarrow{G^{\frac{m}{2}-3}}\left(m-3,\frac{m-4}{2}\right)\xrightarrow{R}\left(m-1,\frac{m}{2}\right)\xrightarrow{G^{\frac{m}{2}-2}}(m-1,m-2)\xrightarrow{A}(0,0).$$

So

$$T_0(m-4)=0,\qquad \rho_0(m-4)=1+\left(\frac{m}{2}-3\right)+1+\left(\frac{m}{2}-2\right)+1=m-2.$$

Case $x=m-3$**.** The orbit is

$$(m-3,0)\xrightarrow{B}(m-2,1)\xrightarrow{G^{m-4}}(m-2,m-3)\xrightarrow{A}(m-1,m-1)\xrightarrow{Q}(2,2)\xrightarrow{G^{m-2}}(2,0).$$

Hence

$$T_0(m-3)=2,\qquad \rho_0(m-3)=1+(m-4)+1+1+(m-2)=2m-3.$$

Case $x=m-2$**.** The orbit is

$$(m-2,0)\xrightarrow{R}(0,2)\xrightarrow{G^{m-3}}(0,m-1)\xrightarrow{R}(2,1)\xrightarrow{A}(3,3)\xrightarrow{G^{\frac{m-4}{2}}}\left(3,\frac{m}{2}+1\right)\xrightarrow{R}\left(5,\frac{m}{2}+3\right)\xrightarrow{G^{\frac{m-6}{2}}}(5,0).$$

Therefore

$$T_0(m-2)=5,$$

and

$$\rho_0(m-2)=1+(m-3)+1+1+\frac{m-4}{2}+1+\frac{m-6}{2}=2m-4.$$

Case $x=m-1$**.** The orbit is

$$(m-1,0)\xrightarrow{G^{\frac{m}{2}-1}}\left(m-1,\frac{m}{2}-1\right)\xrightarrow{R}\left(1,\frac{m}{2}+1\right)\xrightarrow{G^{\frac{m}{2}-1}}(1,0).$$

Hence

$$T_0(m-1)=1,\qquad \rho_0(m-1)=\left(\frac{m}{2}-1\right)+1+\left(\frac{m}{2}-1\right)=m-1.$$

These cases exhaust $x\in\mathbb{Z}_m$. □

Proposition 14 (splicing of arithmetic runs for color 0, Case II)**.** *Assume $m \equiv 4 \pmod 6$. Then T_0 is the successor permutation on a cyclic order obtained from four long arithmetic runs together with $O(1)$ bridge vertices. More precisely, when $m \equiv 10 \pmod{12}$ one has*

$$(0, m-2, 5, 9, \ldots, m-1, 1 \mid 4, 8, \ldots, m-6 \mid 3, 7, \ldots, m-3 \mid 2, 6, \ldots, m-4),$$

while when $m \equiv 4 \pmod{12}$ one has

$$(0, m-2, 5, 9, \ldots, m-3 \mid 2, 6, \ldots, m-6 \mid 3, 7, \ldots, m-1, 1 \mid 4, 8, \ldots, m-4).$$

Consequently T_0 is a single m-cycle.

Proof. Proposition 13 shows that, away from the boundary values

$$0, 1, 2, m-6, m-5, m-4, m-3, m-2, m-1,$$

the map acts by $x \mapsto x+4$. Thus the generic behavior is a disjoint union of long arithmetic runs modulo 4, and the exceptional values determine only how these runs are spliced together.

When $m \equiv 10 \pmod{12}$, the endpoint values give

$$0 \to m-2 \to 5, \qquad m-1 \to 1, \qquad m-6 \to 3, \qquad m-3 \to 2, \qquad m-4 \to 0,$$

while the generic rule $x \mapsto x+4$ propagates along the runs

$$5, 9, \ldots, m-1, \qquad 4, 8, \ldots, m-6, \qquad 3, 7, \ldots, m-3, \qquad 2, 6, \ldots, m-4.$$

This is exactly the first displayed cyclic concatenation.

When $m \equiv 4 \pmod{12}$, the endpoint values give

$$0 \to m-2 \to 5, \qquad m-3 \to 2, \qquad m-6 \to 3, \qquad m-1 \to 1, \qquad m-4 \to 0,$$

and the same generic rule $x \mapsto x+4$ propagates along the runs

$$5, 9, \ldots, m-3, \qquad 2, 6, \ldots, m-6, \qquad 3, 7, \ldots, m-1, \qquad 4, 8, \ldots, m-4.$$

So this time one obtains the second displayed cyclic concatenation.

In either congruence class Lemma 12 applies, and T_0 is a single m-cycle. □

Corollary 8. *If $m \equiv 4 \pmod 6$, then R_0 is a single m^2-cycle.*

Proof. Proposition 14 shows that T_0 is a single m-cycle. Also

$$\sum_{x \in \mathbb{Z}_m} \rho_0(x) = 1 + (m-6)(m-2) + 2(2m-4) + 2(m-1) + (2m-3) = m^2.$$

Lemma 7 finishes the proof. □

Corollary 9. *For every even $m \geq 6$, the return map R_0 is a single m^2-cycle on P_0.*

Proof. Combine Corollaries 7 and 8. □

D An explicit decomposition for $m = 4$

For $m = 4$, define a direction assignment by specifying at each vertex $(i, j, k) \in (\mathbb{Z}_4)^3$ a direction triple $(d_0, d_1, d_2) \in \{0, 1, 2\}^3$, where $0, 1, 2$ correspond respectively to the directions $+i, +j, +k$. The entry in the table for fixed i and coordinates (j, k) is the three-digit word $d_0 d_1 d_2$.

$i = 0$

j \ k	0	1	2	3
0	210	012	120	021
1	201	021	120	210
2	120	012	201	210
3	201	201	210	102

$i = 1$

j \ k	0	1	2	3
0	120	210	120	210
1	102	021	201	012
2	021	201	210	120
3	210	201	012	201

$i = 2$

j \ k	0	1	2	3
0	021	210	201	021
1	012	201	120	210
2	210	120	210	102
3	102	102	012	210

$i = 3$

j \ k	0	1	2	3
0	021	201	012	120
1	210	210	120	021
2	201	021	201	210
3	201	120	201	210

The direction table above is the finite witness. Because $D(4)$ has only 64 vertices, the table alone is sufficient for a complete finite verification: one can follow each color from any starting vertex and check that the orbit has length 64. For convenience, the accompanying file `m4_witness.json` records the direction table together with the three color orbits obtained by starting from $(0, 0, 0)$, and the script `verify_m4_witness.py` automates this finite audit:

1. it checks that every table entry is a permutation of $(0, 1, 2)$;
2. it follows each color from $(0, 0, 0)$ using the table and verifies that the orbit has length 64 before returning;
3. it confirms that the stored machine-readable cycle lists agree with those computed directly from the table.

This keeps the appendix readable while preserving a complete explicit witness for the boundary case $m = 4$.

E Verification artifacts

The verification artifacts focus on the even case $m \geq 6$, which involves the longest case analysis. The odd construction is given by a short closed form in the main text. For the boundary case $m = 4$, the same directory includes the finite witness `m4_witness.json` and its checker `verify_m4_witness.py`.

The main script `route_e_even.py` checks that, for every even m with $6 \leq m \leq 60$:

1. the resulting direction assignment is a valid coloring (each color is a permutation and the three outgoing arcs at every vertex are distinct);
2. the return maps on P_0 agree with the displayed piecewise formulas;
3. the induced return maps on P_0 are single m^2-cycles;

4. each color permutation is a single Hamilton cycle.

Two auxiliary scripts target the most verification-heavy steps in the appendices:

(a) `routee_return_formula_tables_check.py` checks the partition sets, low-layer words, and return-map formulas of Proposition 6;

(b) `routee_first_return_check.py` checks the appendix first-return formulas against the actual return-map dynamics on the stated transversals.

For larger even m, the main script can also validate the construction by checking only the closed-form first-return maps and return-time sums. These computations are not part of the proofs in the present paper; they serve as independent verification of the even-case construction and as regression checks for the displayed closed-form formulas.

Supporting information

Additional supporting information can be found in the supporting-information files submitted with this article. The submission bundle contains the directory `anc/`, including the machine-readable $m = 4$ witness, the Route E verification scripts, and the supporting-information manifest `SuppInfo_manifest.md`.